



Government of Nepal
Ministry of Urban Development (MoUD)
Central Level Project Implementation Unit (CLPIU)
Babarmahal, Kathmandu

Design Catalogue Volume- II

6 June, 2017

Catalogue for Alternative Construction Materials and Technologies

DESIGN CATALOGUE FOR RECONSTRUCTION OF EARTHQUAKE RESISTANT HOUSES

VOLUME II



MARCH, 2017 (FALGUN, 2073)



GOVERNMENT OF NEPAL
MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND BUILDING CONSTRUCTION
BABARMAHAL, KATHMANDU

Design Catalogue Volume-II

Features

- The Catalogue is approach of Nepal Government for Alternative Construction Materials and Technologies for earthquake resilient construction.
- The catalogue include 12 different technology with following information.
 - 3D view of the design
 - Floor plans
 - Elevations
 - Section
 - Structural Details
 - Quantity estimate of major materials
- The design is based on Structural calculations, model test and analytical test and comes under type A and type B building as per NBC.
- Based on principles for housing and human settlements recovery and reconstruction as as indicated in PDNA report.

LIST OF MODELS

S.N.	TECHNOLOGY	MODEL NO.	
1	INTERLOCKING BRICK MASONRY		
1.1	ONE STOREY	I.B.-1.1	
1.2	TWO STOREY	I.B.-1.2	
1.3	ONE STOREY	I.B.-1.3	
2	CONFINED HOLLOW CONCRETE BLOCK MASONRY		
2.1	TWO STOREY	C.H.C.-2.1	
3	HOLLOW CONCRETE BLOCK MASONRY		
3.1	TWO STOREY	H.C.B.-3.1	
4	COMPRESSED STABILIZED EARTH BLOCK MASONRY		
4.1	ONE STOREY	C.S.E.B.-4.1	
4.2	TWO STOREY	C.S.E.B.-4.2	
5	RANDOM RUBBLE MASONRY IN MUD MORTAR WITH GI WIRE CONTAINMENT		
5.1	ONE STOREY	R.R.M.-5.1	
5.2	TWO STOREY	R.R.M. -5.2	
6	BAMBOO AND STONE MASONRY HYBRID STRUCTURE		
6.1	TWO STOREY	B.S.M.H.-6.1	
7	RAT TRAP BOND MASONRY		
7.1	ONE STOREY	R.T.-7.1	
8	EARTH BAG MASONRY		
8.1	ONE STOREY	E.B.-8.1	
9	LIGHT GAUGE STEEL STRUCTURE		
9.1	ONE STOREY	L.G.S.-9.1	
9.2	TWO STOREY	L.G.S.-9.2	
10	STEEL STRUCTURE		
10.1	TWO STOREY	S.S.-10.1	
11	TIMBER STRUCTURE		
11.1	TWO STOREY	T.S.-11.1	
12	DEBRIS BLOCK MASONRY		
12.1	ONE STOREY	D.B.-12.1	172

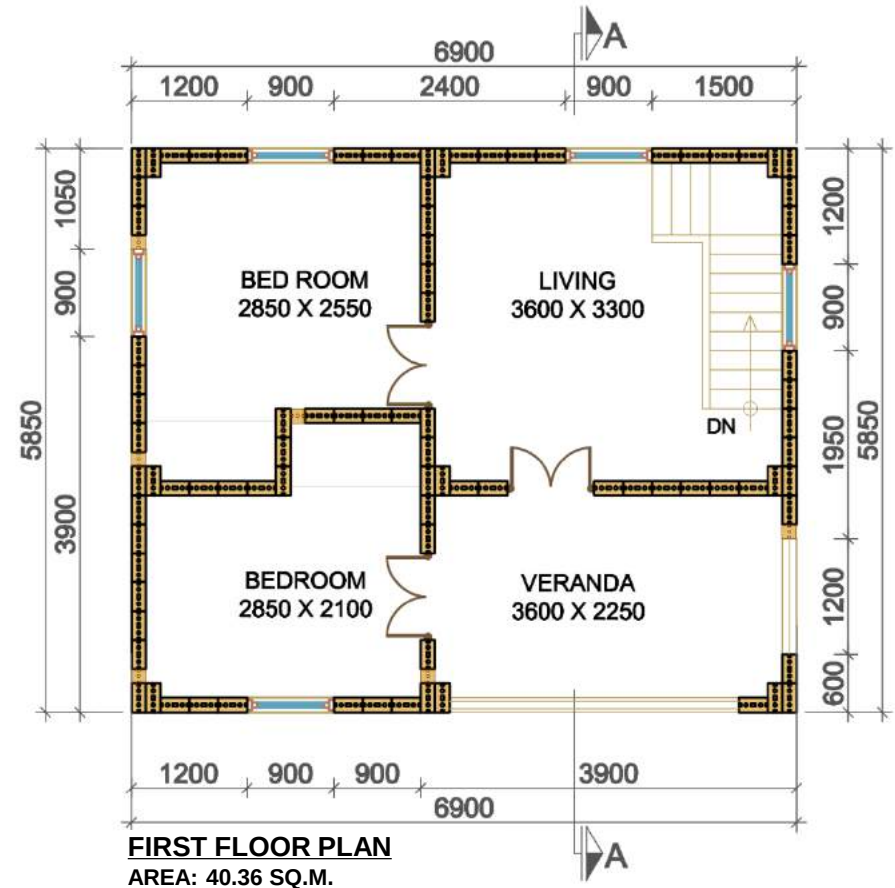
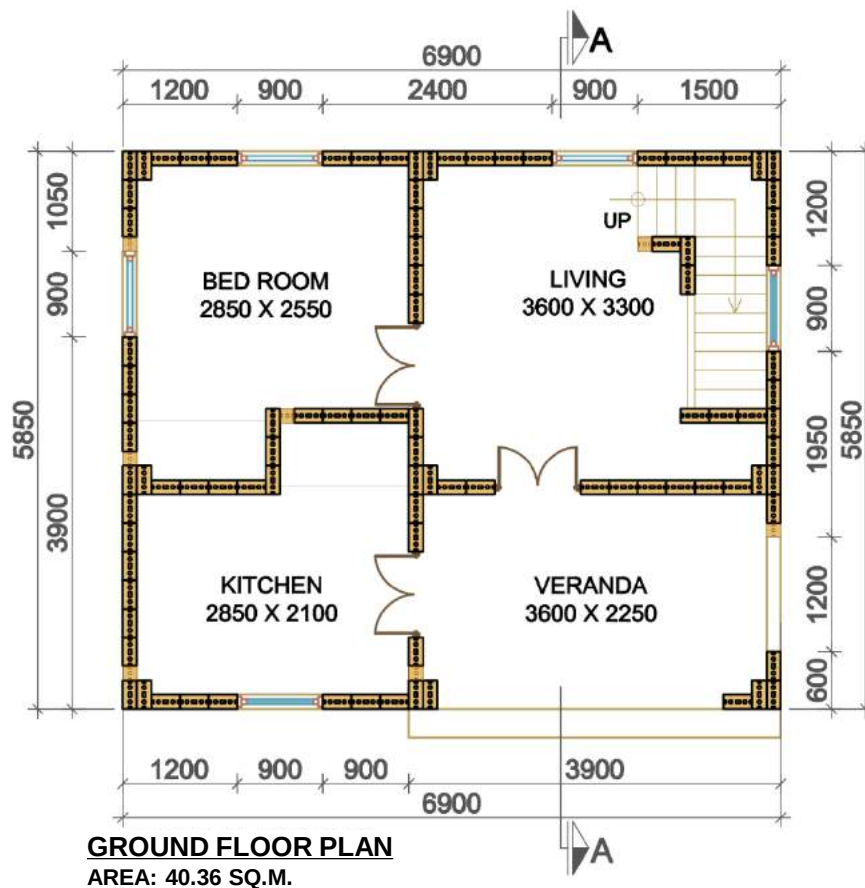
1. INTERLOCKING BRICK MASONRY

- Interlock Brick is made of Laterite Soil (Clay, Silt, Sand and chips) with the certain percentage of Stabilizer (cement).
- The brick is designed with tongue and groove features that allows bricks to interlock each other in masonry wall.
- There is provision of U-shaped brick in Sill and Lintel Level
- The hole with vertical reinforcement is grouted with Micro concrete 1:2:3 (Cement, Sand and Chips) ratio.
- Three model houses I.B 1.1, I.B 1.2, I.B 1.3 are the presented in the catalogue.



LEVEL	MATERIALS								
	Stone	Interlocking Bricks	Cement	Sand	Aggregate	Reinforcing Bar	CGI sheet	GI Sheet	Wood
	Cu.m	No.	Bags	Cu.m.	Cu.m.	Kg.	Bundle	Sq.m.	Cu.m.
Up to Plinth Level	18.1	-	80.5	10.9	6.1	177.2			-
Super Structure	-	6,447.0	57.5	3.0	6.1	493.9			1.3
Roofing	-	-	-	-	-	-	4.2	8.7	1.5
TOTAL	18.1	6,447.0	138	13.9	12.2	671.1	4.2	8.7	2.8







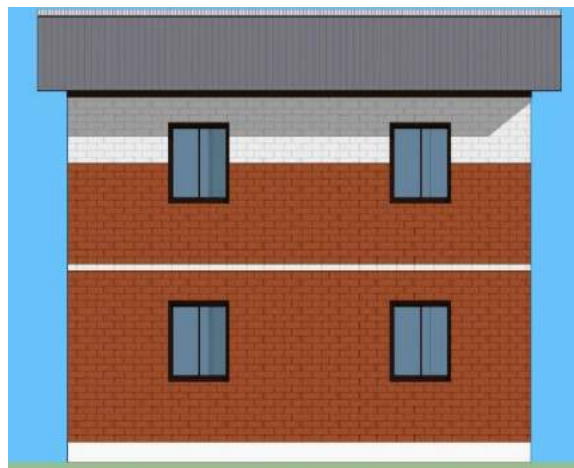
SIDE ELEVATION



FRONT ELEVATION



SIDE ELEVATION

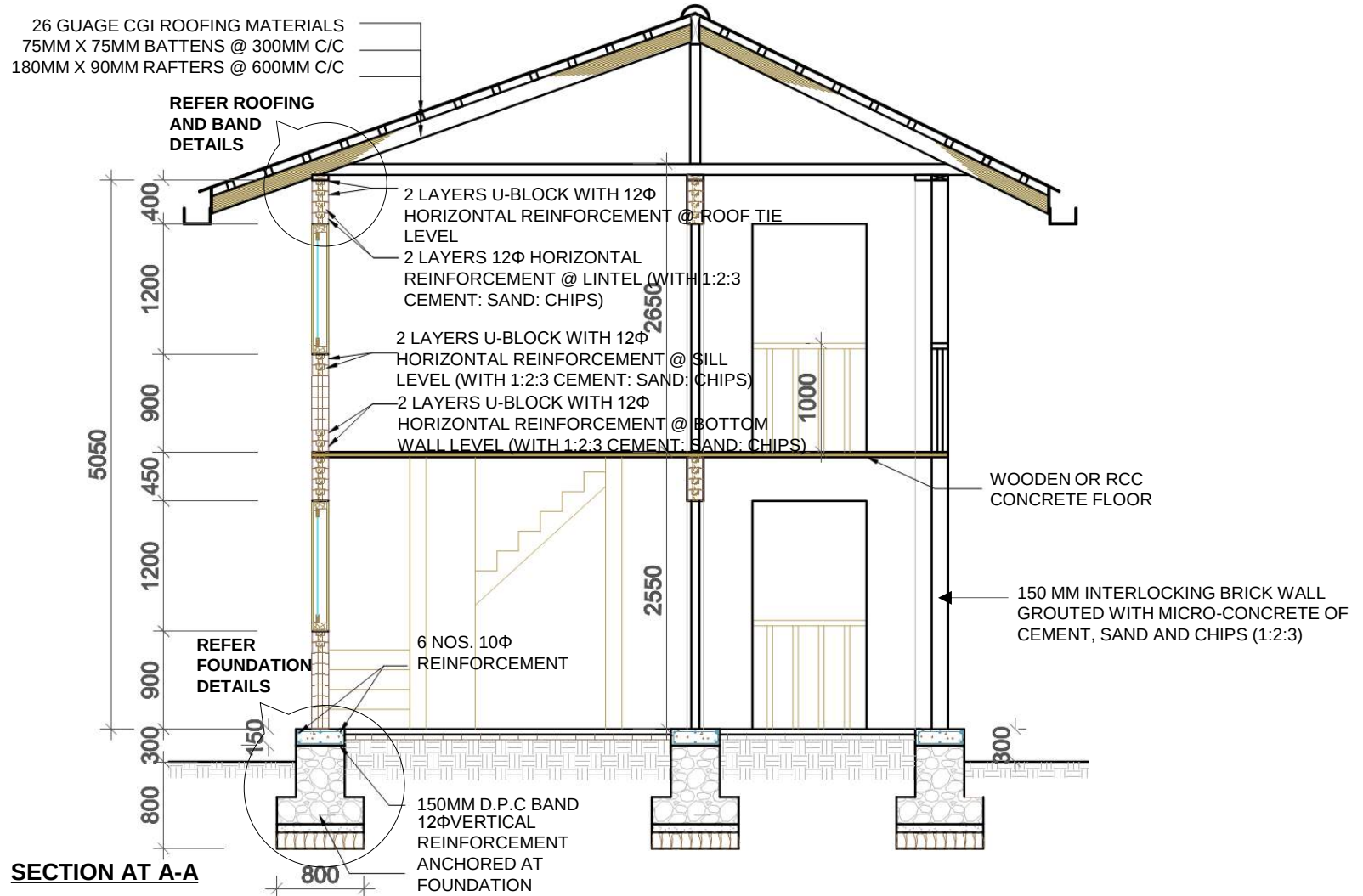


BACK ELEVATION



MODEL I.B.-1.2, INTERLOCKING BRICK MASONRY

TWO STOREY



MINISTRY OF URBAN DEVELOPMENT
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BUILDING CONSTRUCTION

HOUSING TYPE: SECTION

DRAWING TITLE: 11

SCALE: NONE

DATE:

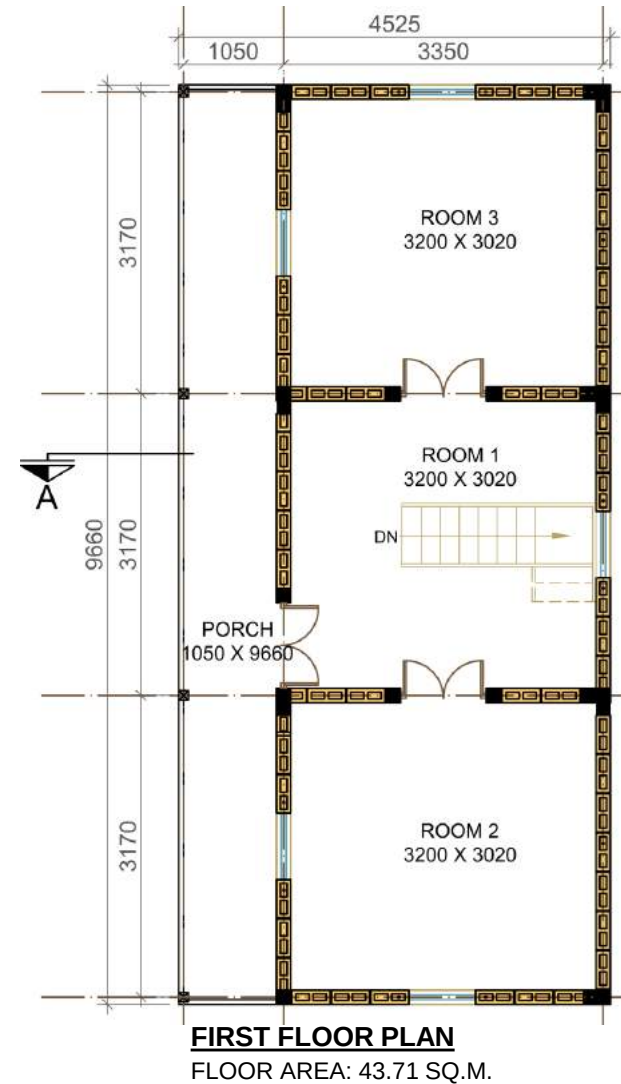
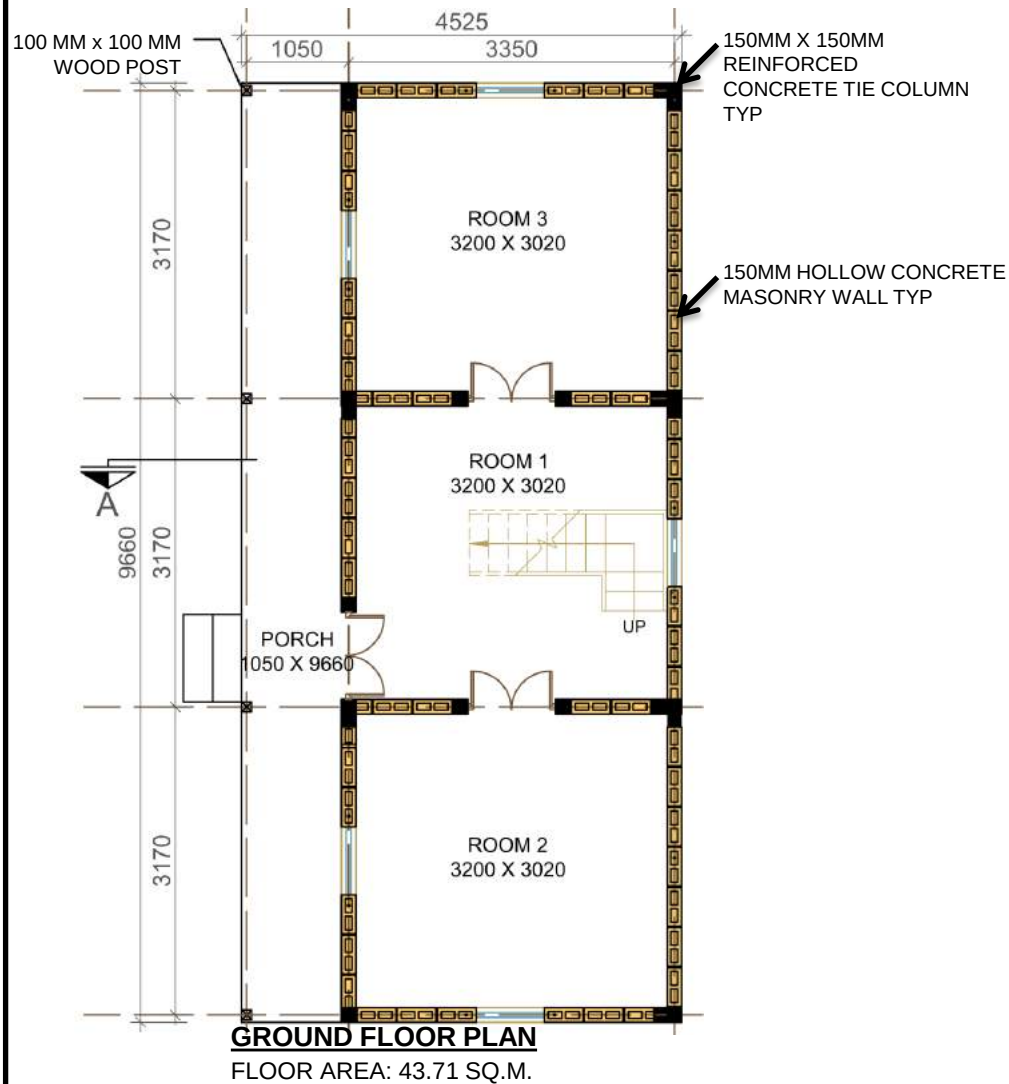
2. CONFINED HOLLOW CONCRETE BLOCK MASONRY

- Hollow concrete block having compressive strength greater than 5Mpa is used as masonry units.
- Tie Column and Tie beam is constructed in sequences with the masonry wall.
- .Hollow concrete block walls with toothing are constructed up to sill level leaving space for columns and then columns and sill are monolithically casted.
- The catalogue included the model C.H.C.-2.1 with two storied structure and six rooms.



LEVEL	MATERIALS								
	Stone	Hollow Concrete Bricks	Cement	Sand	Aggregate	Reinforcing Bar	CGI sheet	GI Sheet	Wood
	Cu.m	No.	Bags	Cu.m.	Cu.m.	Kg.	Bundle	Sq.m.	Cu.m.
Up to Plinth Level	30.3	-	78	12.3	3.5	252.0			-
Super Structure	-	1,330.0	102	8.7	7.5	1,388.0			1.5
Roofing	-						5.0	6.5	1.7
TOTAL	30.3	1,330.0	180	21.0	11.0	1,640.0	5.0	6.5	3.2



MODEL C.H.C.-2.1, CONFINED HOLLOW CONCRETE BLOCK MASONRY**TWO STOREY**

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BUILDING CONSTRUCTION

HOUSING TYPE: **FLOOR PLANS**

DRAWING TITLE: 30

SCALE: NONE

DATE:



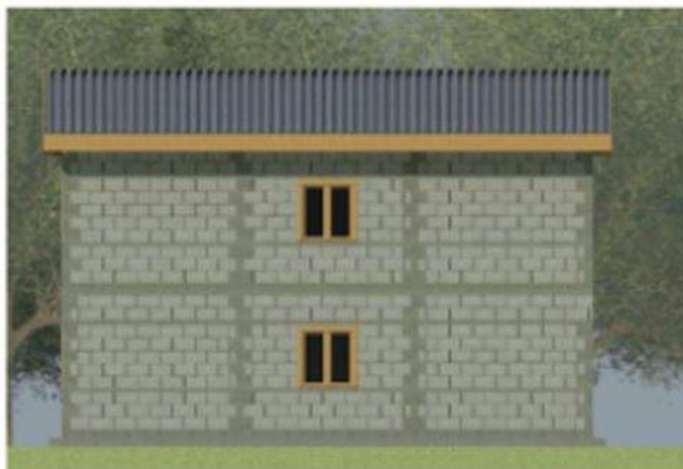
FRONT ELEVATION



SIDE ELEVATION

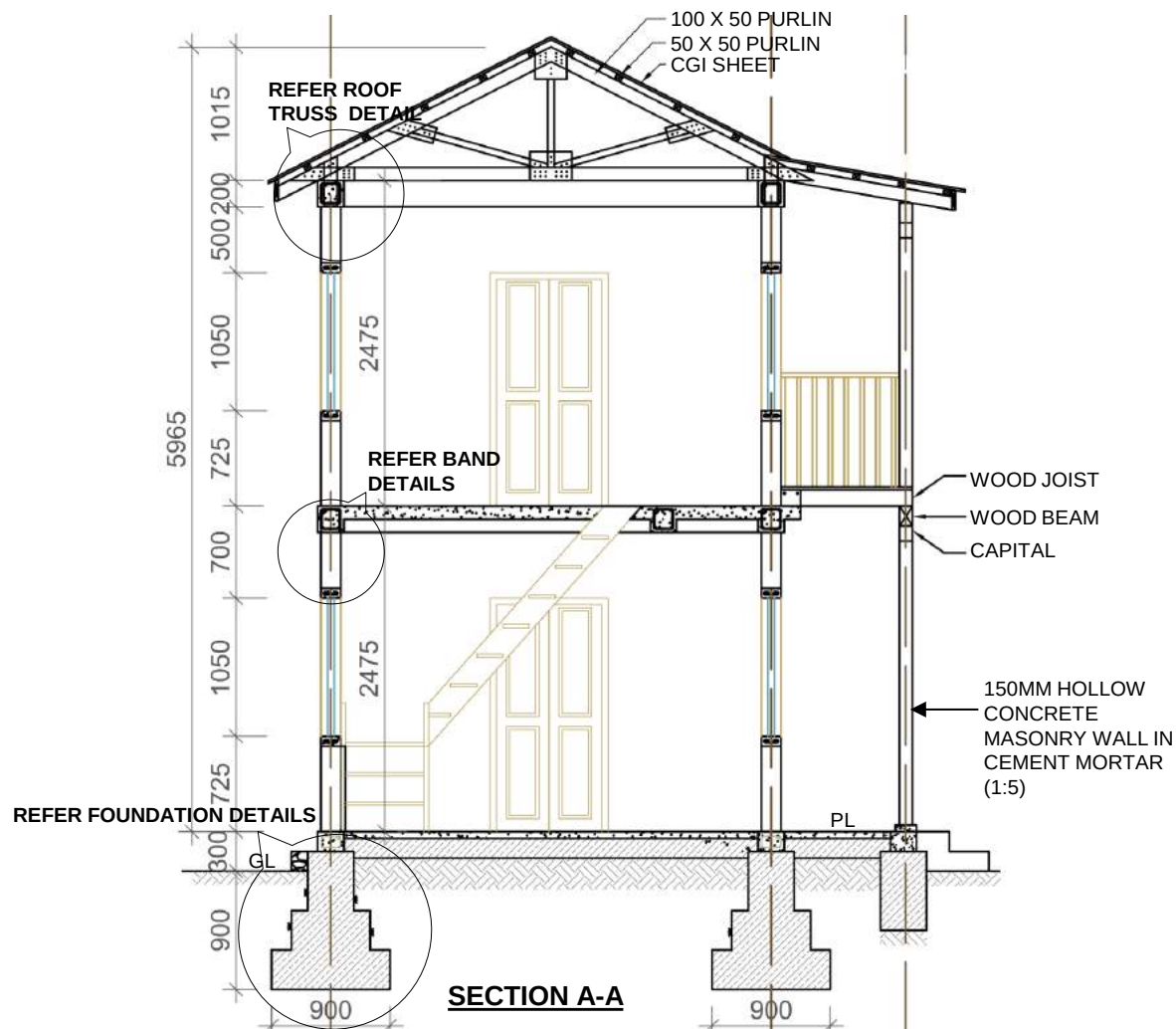


SIDE ELEVATION



BACK ELEVATION





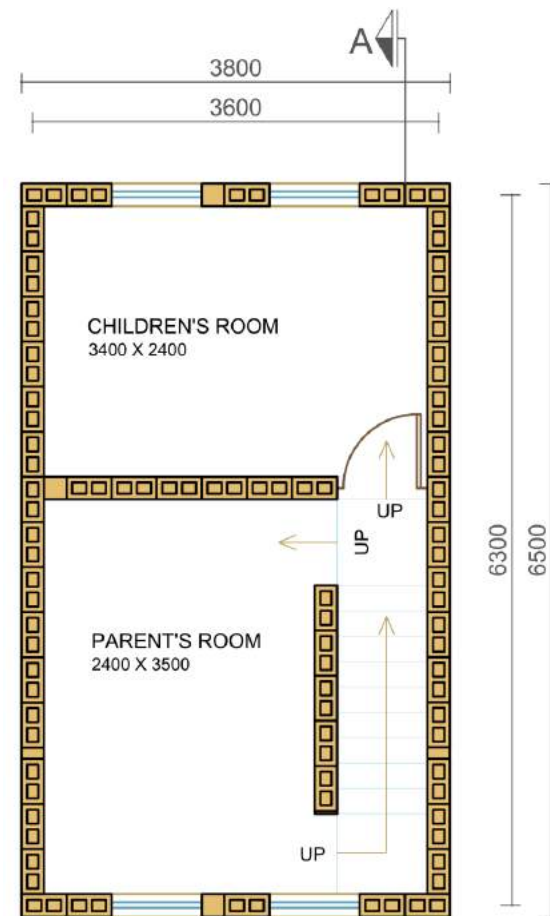
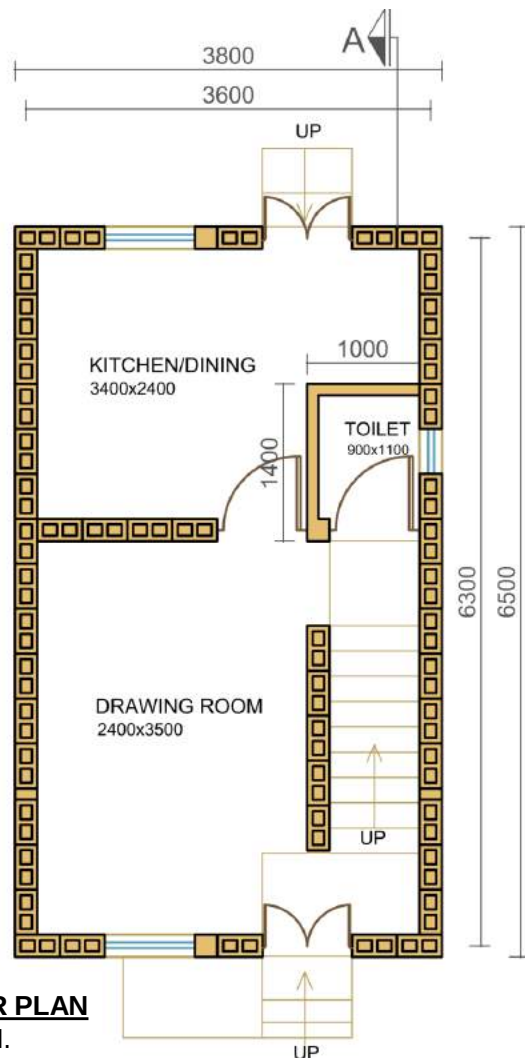
HOLLOW CONCRETE BLOCK MASONRY

- The hollow concrete block masonry is load bearing structure and is alternative to burnt brick masonry.
- The block size used is 40cm X 20cm X10cm with Compressive strength greater than 5 Mpa.
- Design features are RCC strip foundation, load bearing hollow concrete walls, precast floor and roof, precast stair slabs, horizontal bands and vertical seismic reinforcement at critical sections
- H.C.B. 3.1 is a two storied residence with four rooms.



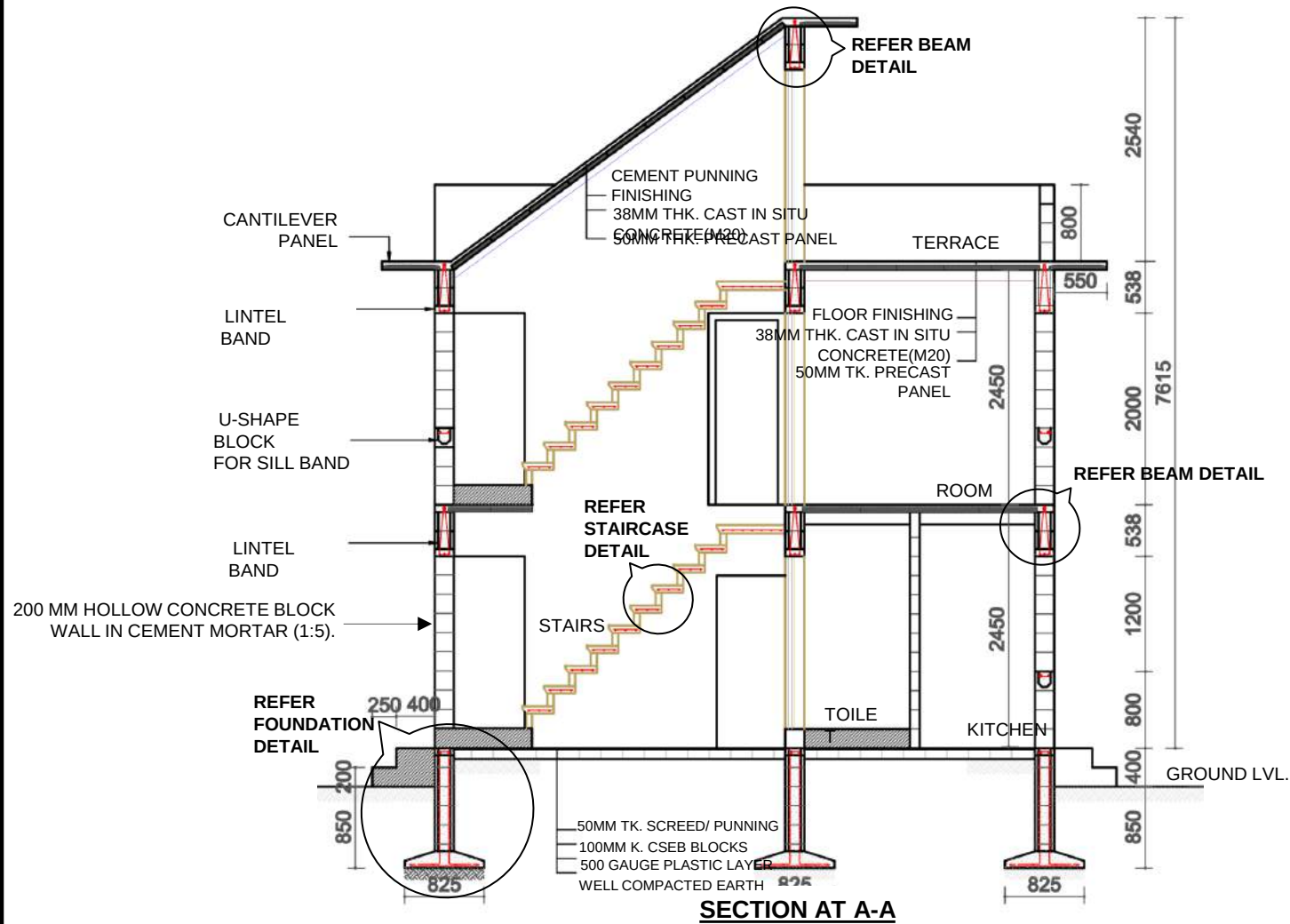
LEVEL	MATERIALS					
	Hollow Concrete Bricks	Cement	Sand	Aggregate	Reinforcing Bar	Polythene sheet
	No.	Bags	Cu.m.	Cu.m.	Kg.	Sq.m
Up to Plinth Level	447.0	100	6.1	12.1	864.6	19.7
Super Structure	2,398.0	163	8.6	17.5	1,677.3	
Roofing						
TOTAL	2,845.0	263	14.7	29.6	2,541.9	19.7





TWO STOREY

TWO STOREY

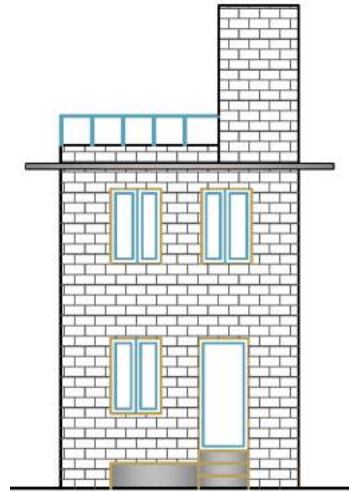


HOUSING TYPE: SECTION

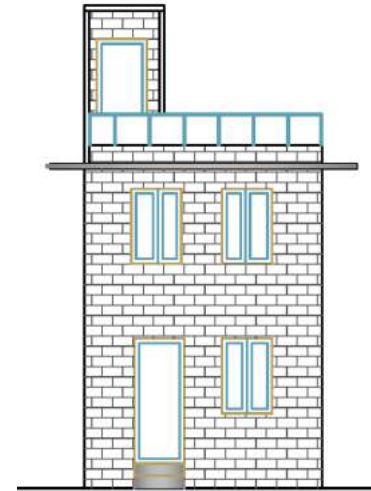
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DRAWING TITLE: 44

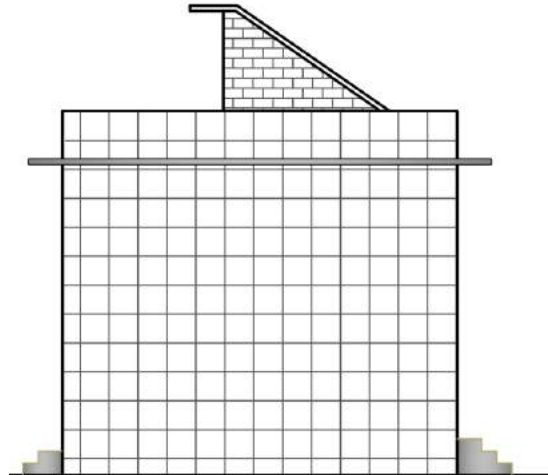
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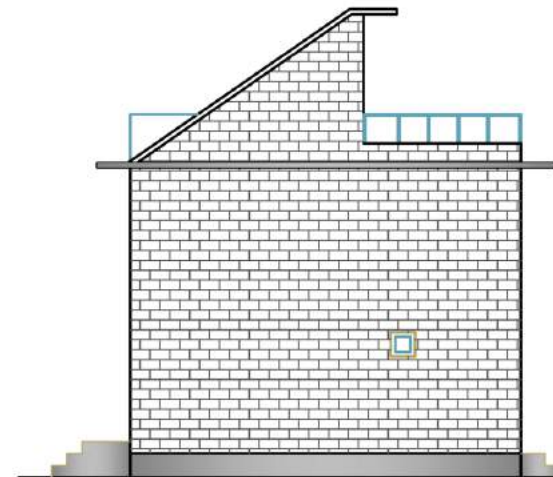
FRONT ELEVATION



BACK ELEVATION



SIDE ELEVATION



SIDE ELEVATION



4. COMPRESSED STABILIZED EARTH BLOCK MASONRY

- Compressed Stabilized Earth Block (**CSEB**) Technology makes use of mud as a predominant building material with promotion of local construction material.
- The properties of soil used are improved by using stabilizers like cement or by chemical such Flat Plug Resin.
- The house of the technology are low cost and is suitable for rural areas.
- Two models C.S.E.B- 4.1 with one storey and C.S.E.B.-4.2 is a two storied residential is included in catalogue.



LEVEL	MATERIALS					
	Stabilized block	Cement	Sand	Reinforcing Bar	Wood	Bamboo
	No.	Bags	Cu.m.	Kg.	Cu.m.	Nos
Up to Plinth Level	1,758.0	17.0	2.3	-	-	-
Super Structure	1,500.0	8	1.4	237.3	0.2	26
Roofing	-	-	-	-	2.4	31.0
TOTAL	3,258.0	25	3.7	237.3	2.6	57

CSEB-4.1

CSEB-4.1



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DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

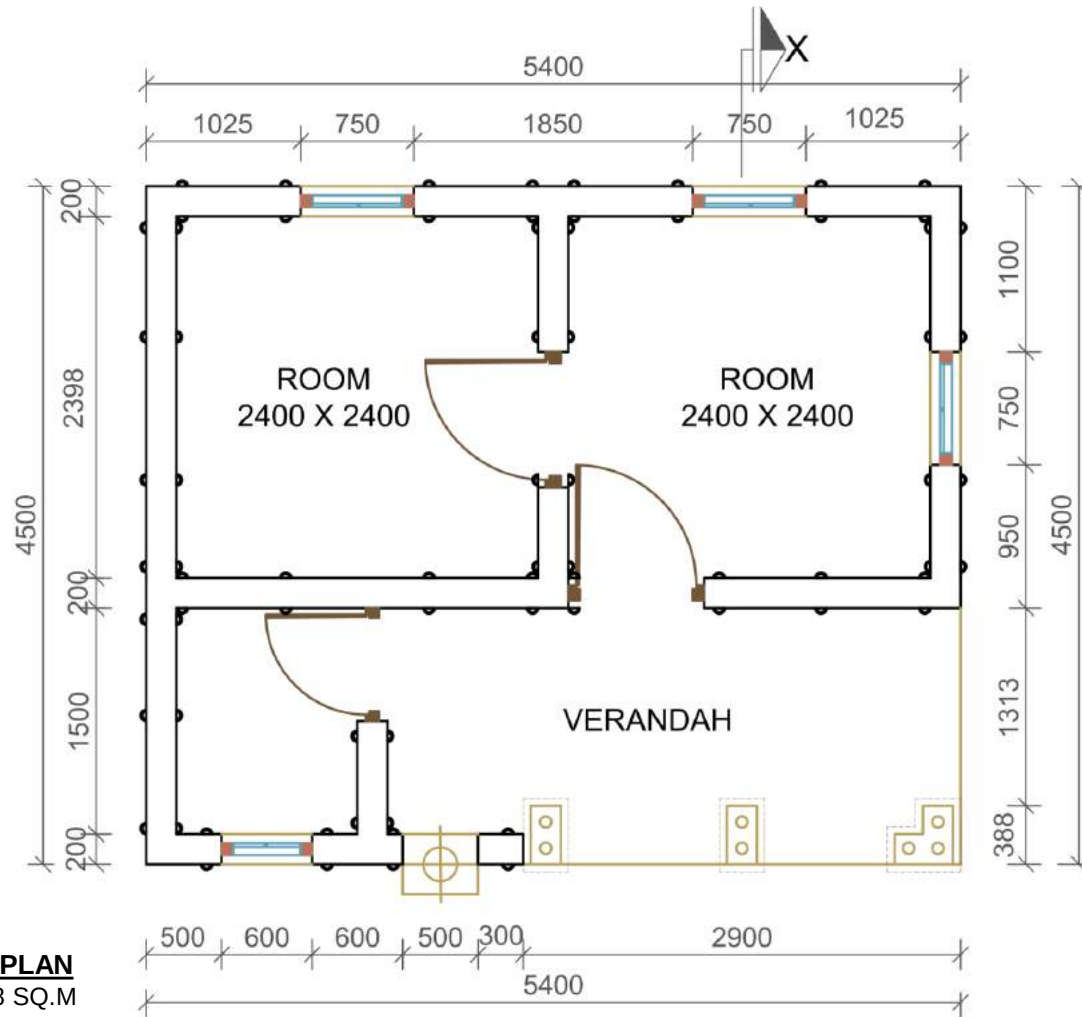
HOUSING TYPE: ESTIMATE AND 3D-VIEW

DRAWING TITLE: 55

SCALE: NONE

DATE:

1/8



GROUND FLOOR PLAN
FLOOR AREA: 24.08 SQ.M

CSEB-4.1

CSEB-4.1

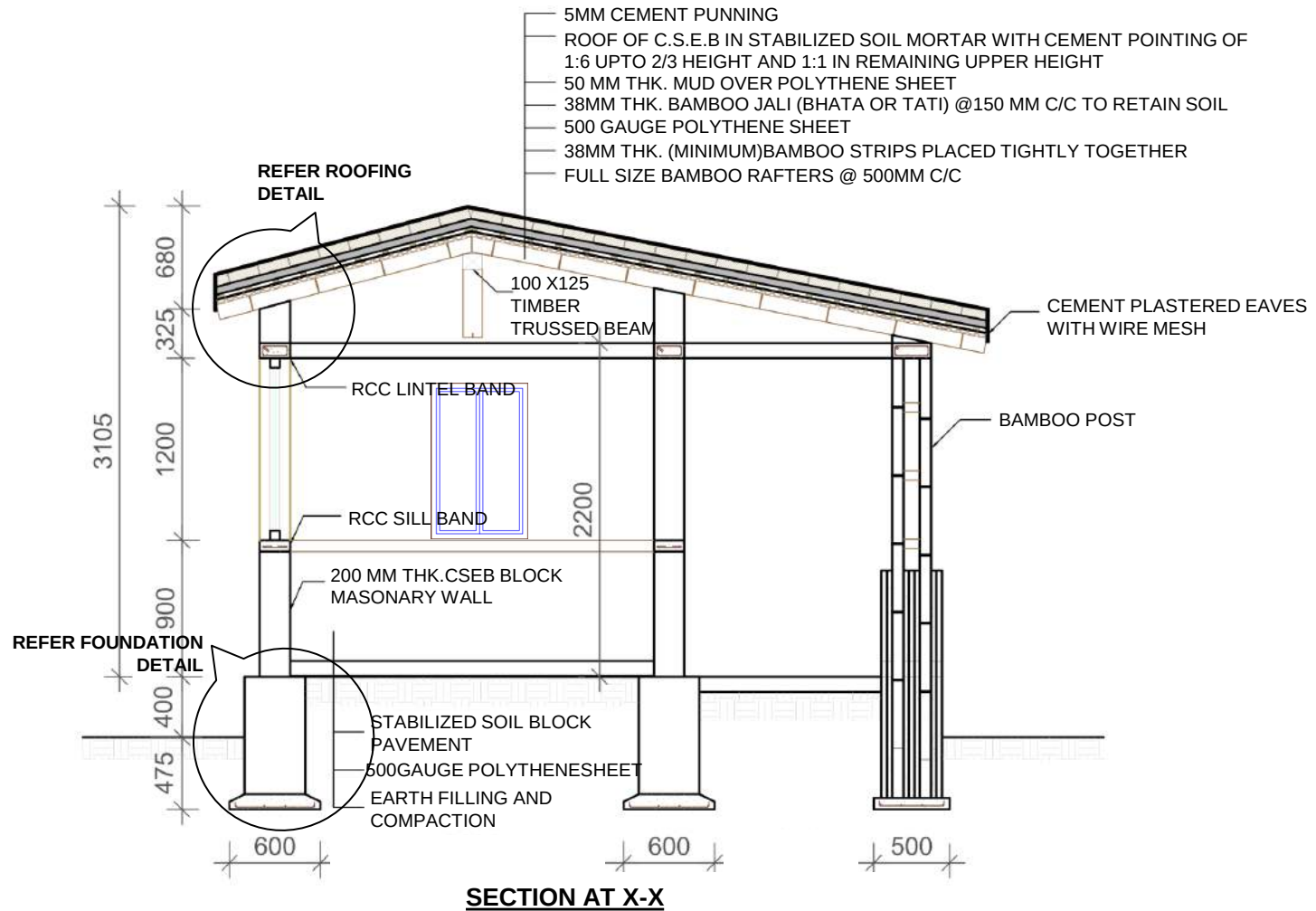


MINISTRY OF URBAN DEVELOPMENT
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BUILDING CONSTRUCTION

HOUSING TYPE: **GROUND FLOOR PLAN**
DRAWING TITLE: 56

SCALE: NONE
DATE:

2/8



CSEB-4.1

CSEB-4.1

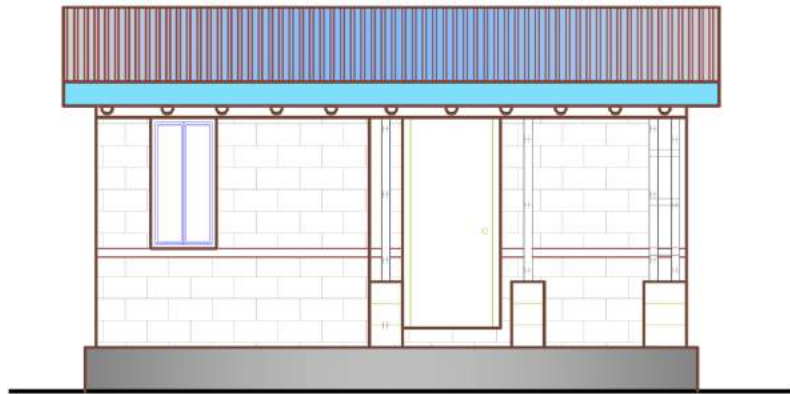


MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

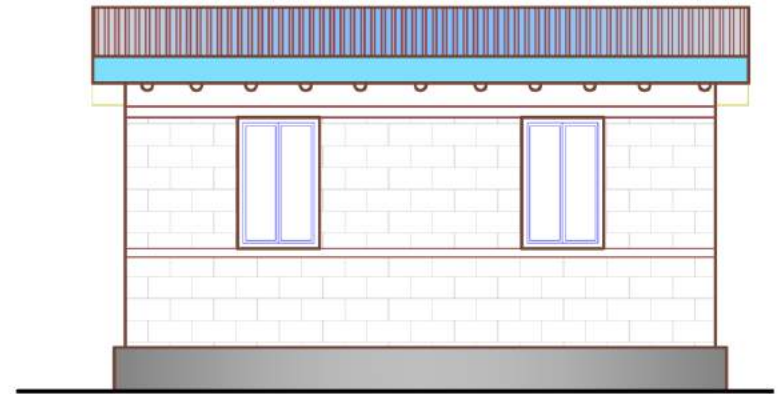
HOUSING TYPE: SECTION
DRAWING TITLE: 57

SCALE: NONE
DATE:

3/8



FRONT ELEVATION



BACK ELEVATION



LEFT ELEVATION



RIGHT ELEVATION

CSEB-4.1

CSEB-4.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: ELEVATIONS

DRAWING TITLE: 58

SCALE: NONE

DATE:

4/8

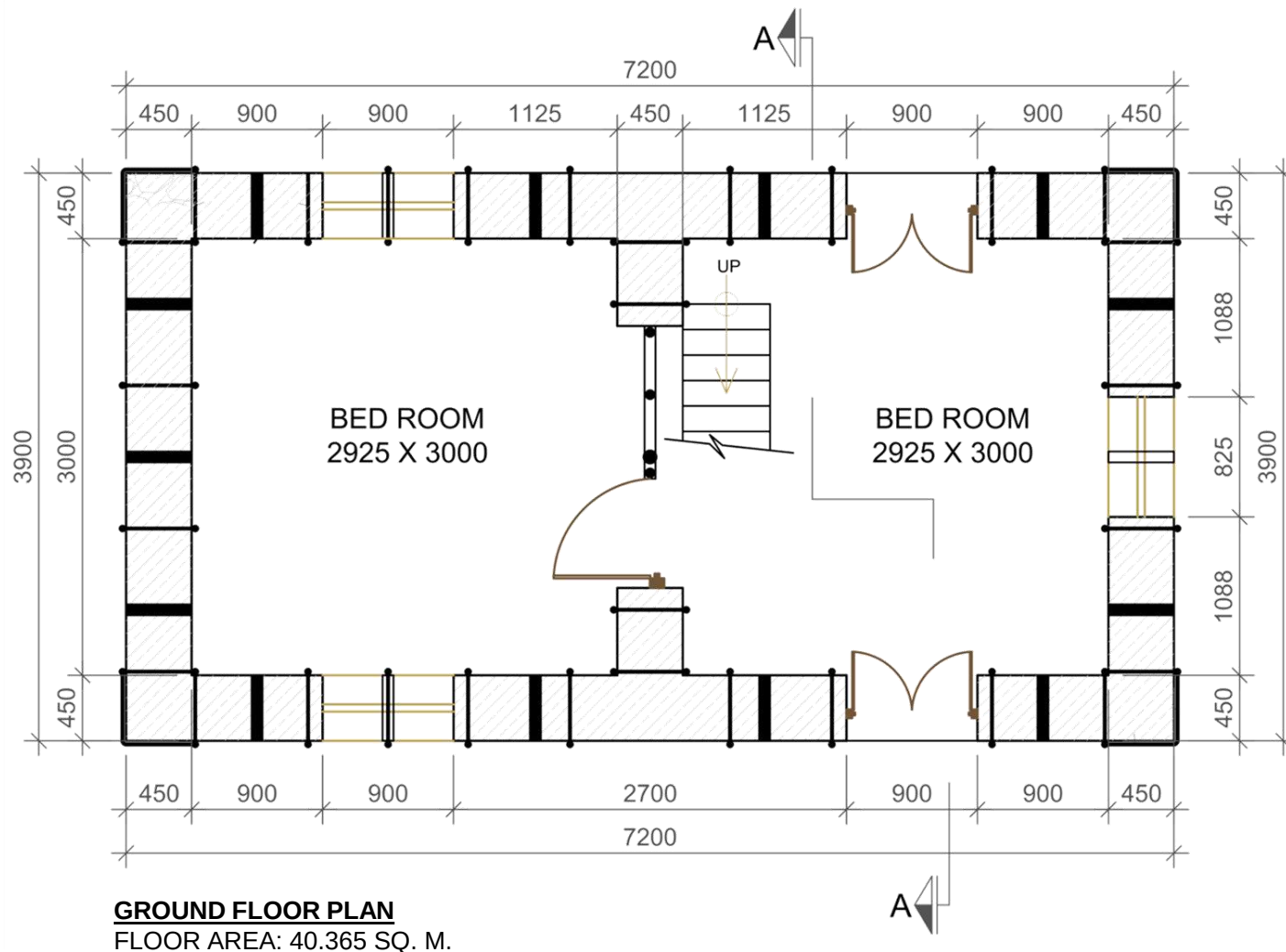
5. RANDOM RUBBLE MASONRY IN MUD MORTAR WITH GI WIRE CONTAINMENT

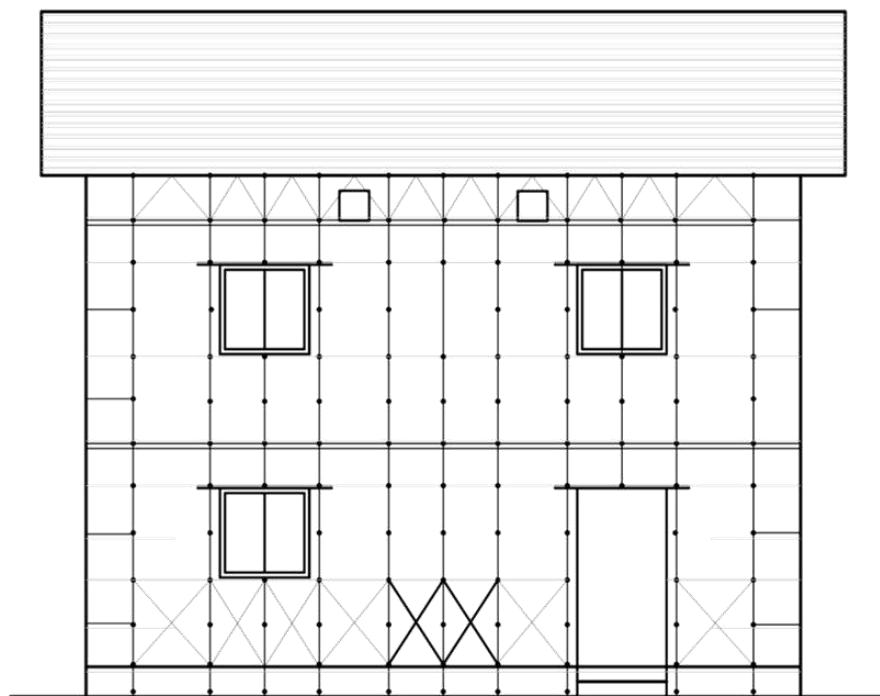
- This technology is an improvement on random rubble masonry structure by introduction of GI containment wires.
- Vertical GI Containment wires are provided on two faces connected by ties of a masonry wall to prevent flexural failure.
- R.R.M. 5.1 is a one storied and R.R.M. 5.2 is a two storied four room unit with CGI sheet roofing is presented in catalogue.
- The basic shape and size of the building comply Nepal National Building Code, NBC 203 : 1994, Guidelines for earthquake resistant building construction: low strength masonry.



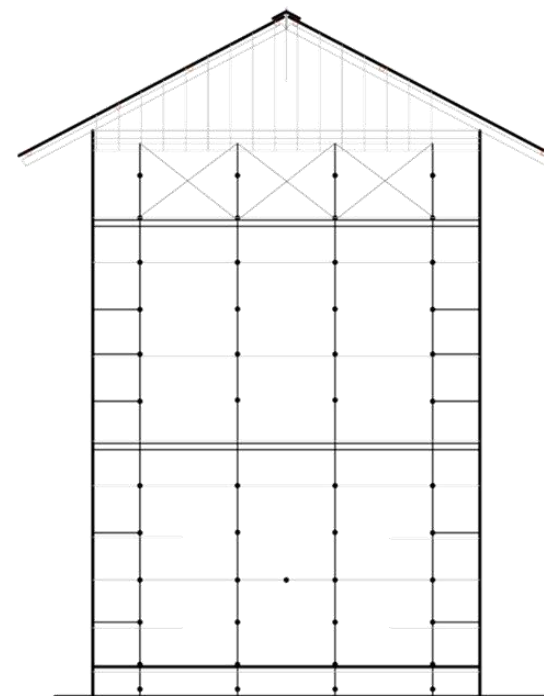
LEVEL	MATERIALS								
	Stone	Weld wire mesh	Sand	Mud	CGI Sheet	GI Sheet	Wood	4mm GI wire	2 mm GI Wire
	Cu.m.	Sq.m	Cu.m.	Cu.m.	Bundle	Sq.m.	Cu.m.	Kg	Kg
Up to Plinth Level	16.1	10.4	1.3	7.4			-	6.0	14.0
Super Structure	49.3	71.9	-	18.8			0.9	28.0	66.0
Roofing	-	-	-	-	4.1	9.1	6.6	-	4.0
TOTAL	65.4	82.3	1.3	26.3	4.1	9.1	7.5	34.0	84.0







FRONT SIDE ELEVATION



LEFT SIDE ELEVATION



BAMBOO AND STONE MASONRY HYBRID STRUCTURE

- The proposed technology uses traditional, yet earthquake resistant construction using materials and skills that are indigenous and locally available. Local bamboo (*Banbusa Nutans*), ***seasoned and treated, is used in a structural*** frame with bamboo wattle and daub panels as walls on the upper floor. The frame is surrounded with a wall in Stone Masonry with Mud Mortar on the ground floor of the house.
- Featured design in H.B.S.M.-5.1 consists of a Ground Floor space that can be converted into two rooms using a lightweight Wattle and Daub partition. A Kitchen and a Covered Verandah flank the room on the short and the long side respectively.

MATERIAL PROPERTIES

Bamboo Properties

Min Compressive Strength of bamboo: 45.6 Mpa

Density of bamboo : 673 Kg/m³

Modulus of elasticity: 10.72 x 10³ Mpa



LEVEL	MATERIALS					
	Stone	Mud	CGI Sheet	GI Sheet	Wood	Bamboo
	Cu.m.	Cu.m.	Bundle	Sq.m.	Cu.m.	Nos
Up to Plinth Level	11.6	13.0			-	230.0
Super Structure	21.3	8.1			0.6	
Roofing	-	-	5.6	9.5	0.7	
TOTAL	32.9	21.1	5.6	9.5	1.3	230.0

B S M H - 6.1



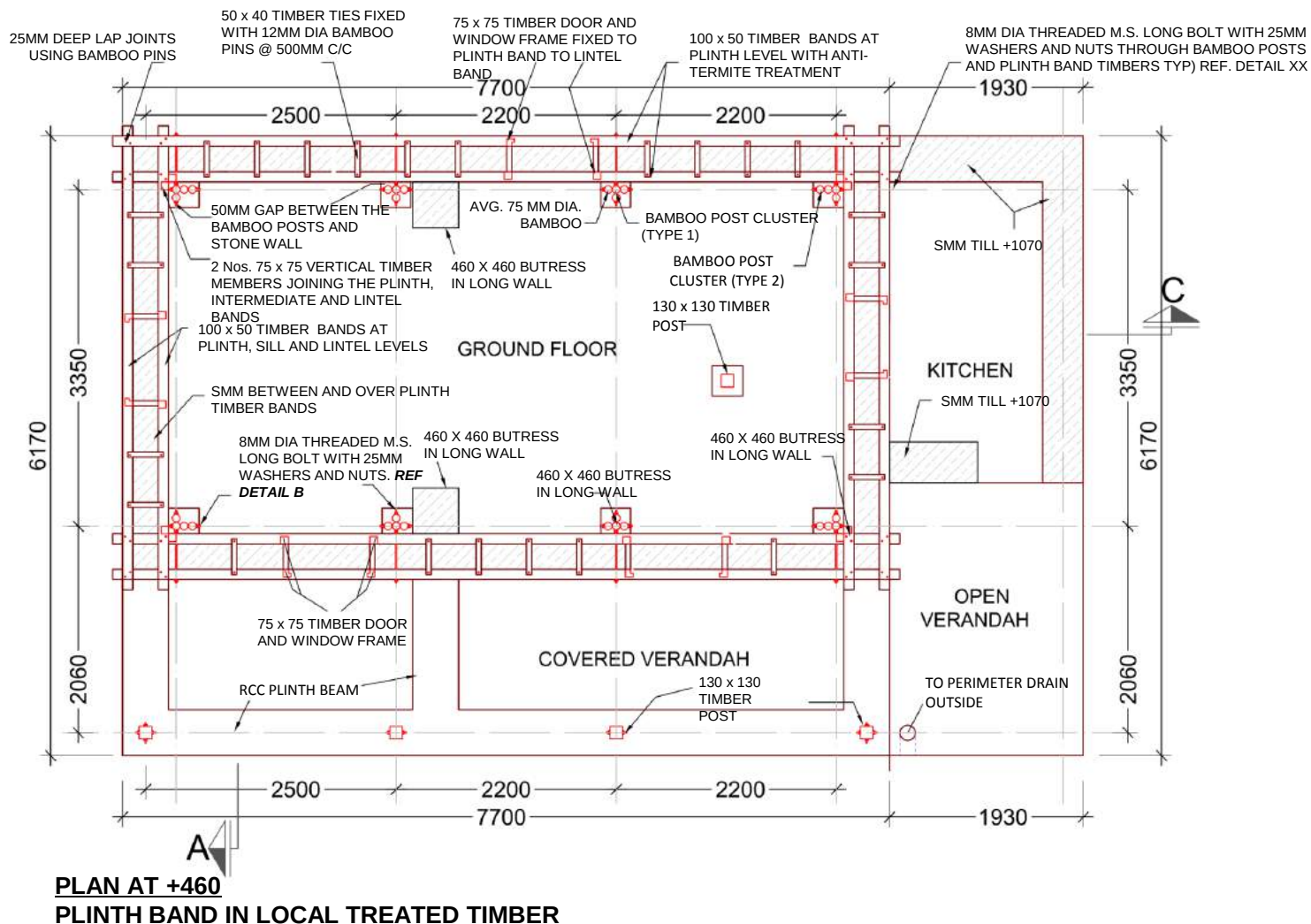
MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: ESTIMATE AND 3D-VIEW
DRAWING TITLE: 6.1

SCALE: NONE
DATE:

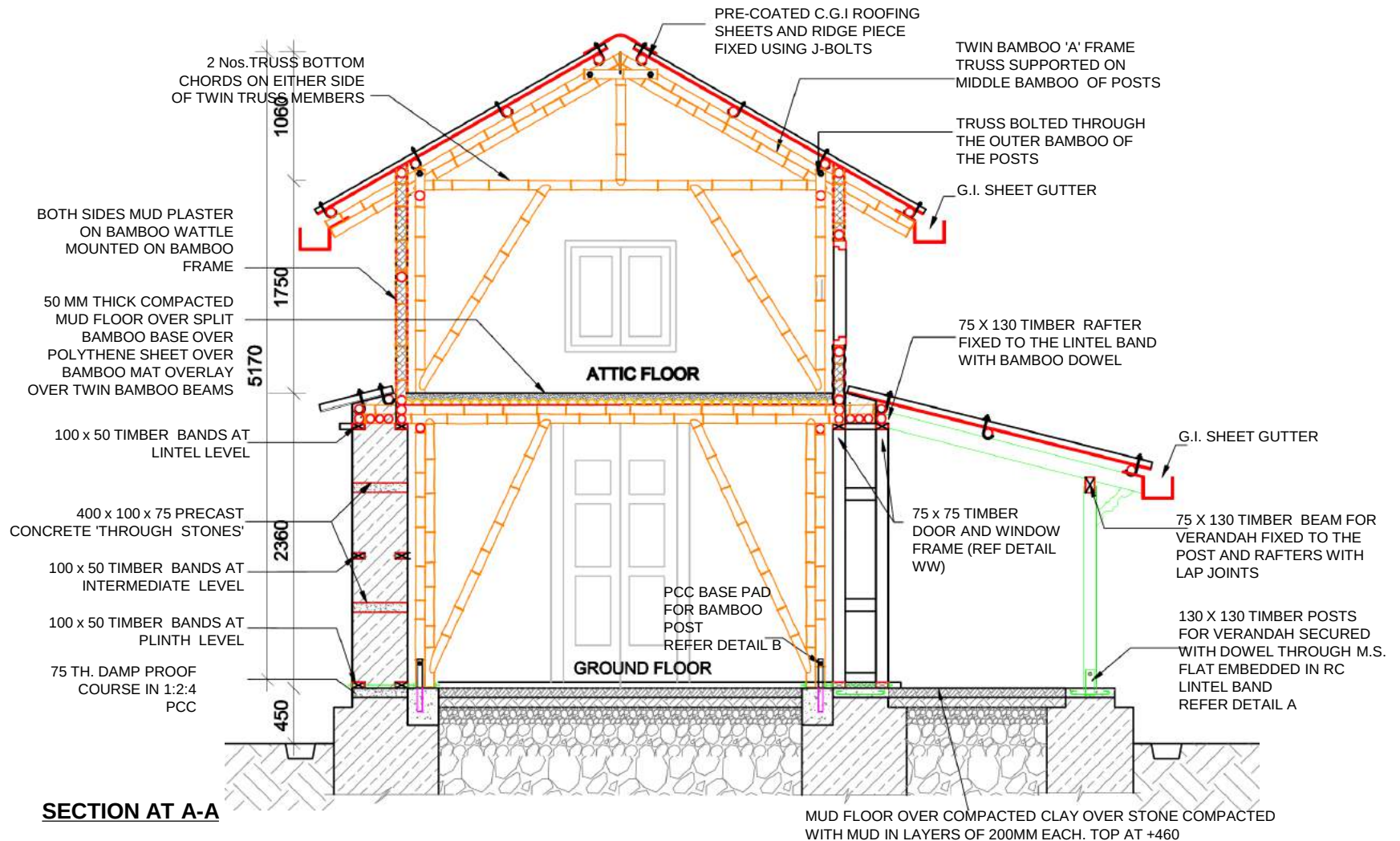
BSMH-6.1

1/14



MODEL B.S.M.H.-6.1. BAMBOO AND STONE MASONRY HYBRID STRUCTURE

TWO STOREY



B.S.M.H.-6.1

BSMH-6.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: SECTION
DRAWING TITLE: 100

SCALE: NONE
DATE:

9/14

7.RAT TRAP BOND MASONRY

- Rat-Trap Bond is a modular type of masonry construction in which bricks are laid on edge, thereby creating an internal cavity within the wall. The cavity improves the thermal behavior of the wall and significantly reduces the quantity of brick and mortar in the masonry. It is a Green Building technology and an appropriate option against conventional solid brick wall masonry from sustainable point of view. Rat trap bond masonry can be used both for partition wall or as a load bearing structures. As Rat trap bond construction is a modular type of masonry construction, due care must be taken while designing the wall length and height.
- The design featured in Model RTB-2.1 is a two roomed single storied load bearing structure of Rat Trap bond masonry. Horizontal bands, vertical reinforcements, corner reinforcement and reinforcement in T- junctions are provided.
- **MATERIAL PROPERTIES**
- Min Compressive Strength of Rat Trap Bond : 1.3 Mpa
- Unit weight of RTB masonry: 15KN/m³
- Young's Modulus: 715 Mpa



LEVEL	MATERIALS								
	Stone	Brick	Cement	Sand	Aggregate	Wood	Reinforcing Bar	CGI Sheet	GI Sheet
	Cu.m.	No.	Bags	Cu.m.	Cu.m.	Cu.m.	Kg.	Bundle	Rm.
Up to Plinth Level	19.5	2,225.0	59.0	10.3	2.5		280.4		
Super Structure		5,125.0	32.0	3.1	2.5	0.6	297.8		
Roofing		-	-	-	-	4.0	-	4.2	10.0
TOTAL	19.5	7,350.0	91.0	13.4	5	4.6	578.2	4.2	10.0

RTB-7.1

RTB-7.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

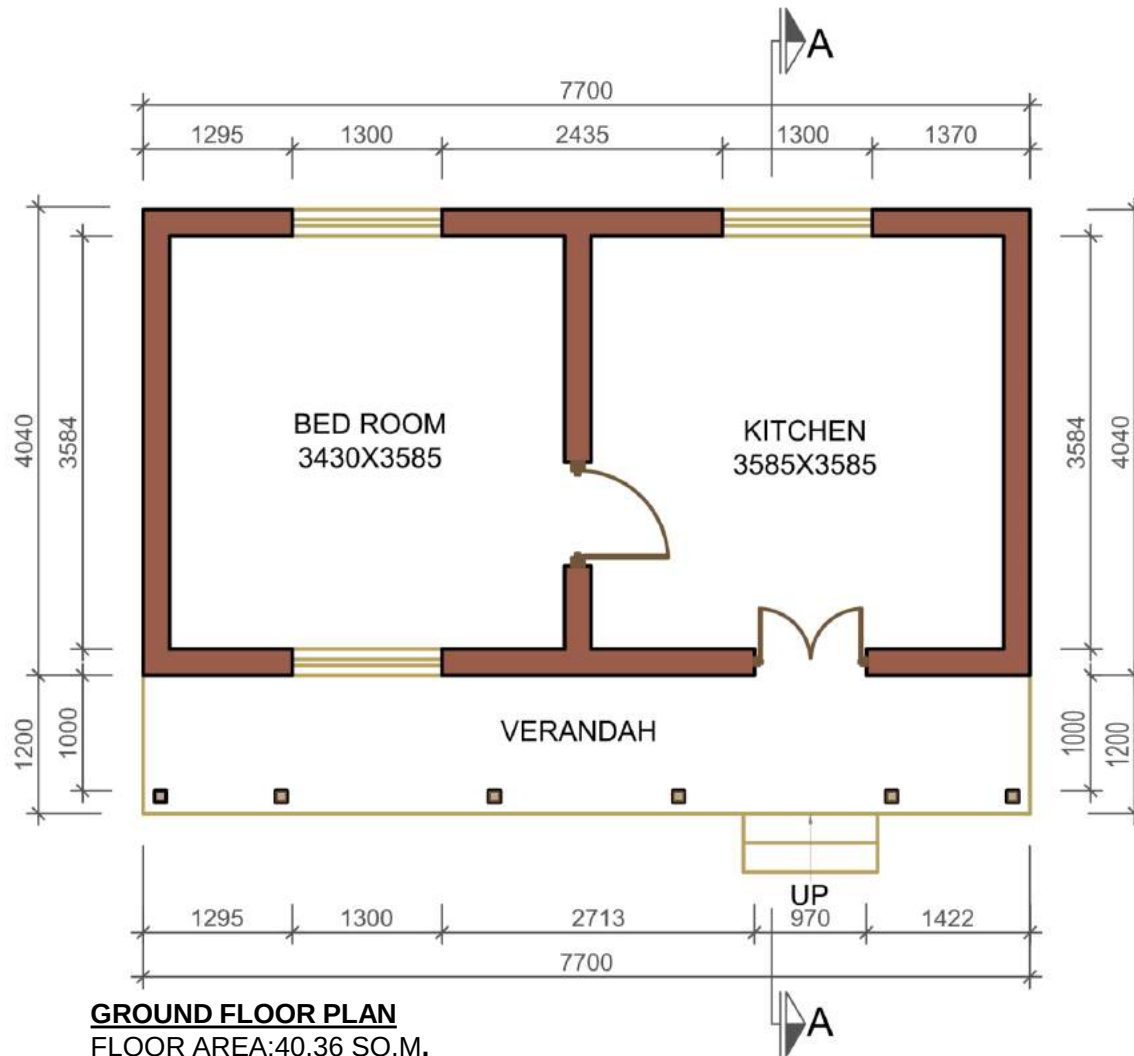
HOUSING TYPE: ESTIMATE AND 3D-VIEW

DRAWING TITLE: 108

SCALE: NONE

DATE:

1/10



RTB-7.1

RTB-7.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

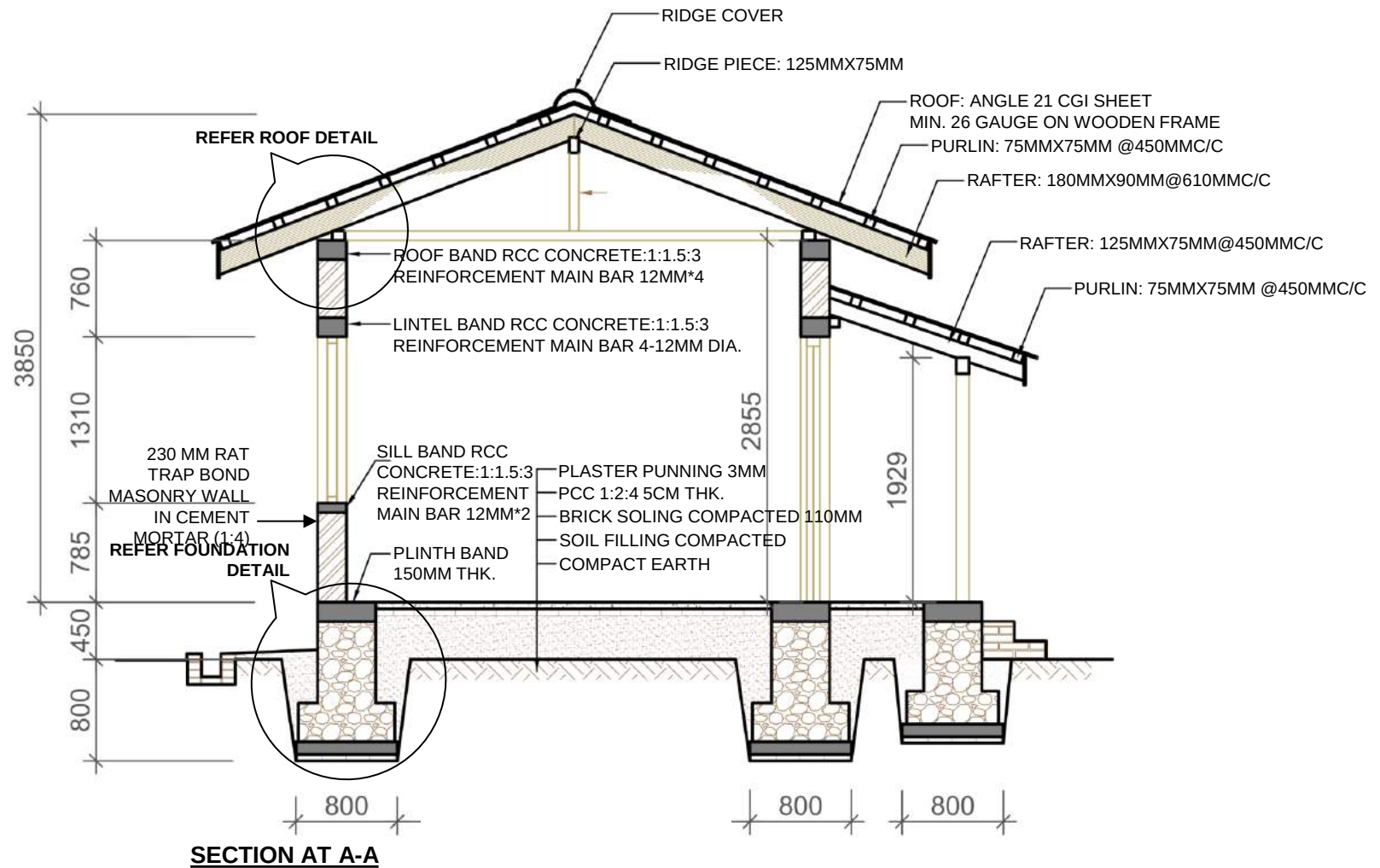
HOUSING TYPE: **GROUND FLOOR PLAN**

DRAWING TITLE: 109

SCALE: NONE

DATE:

2/10



RTB-7.1

RTB-7.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

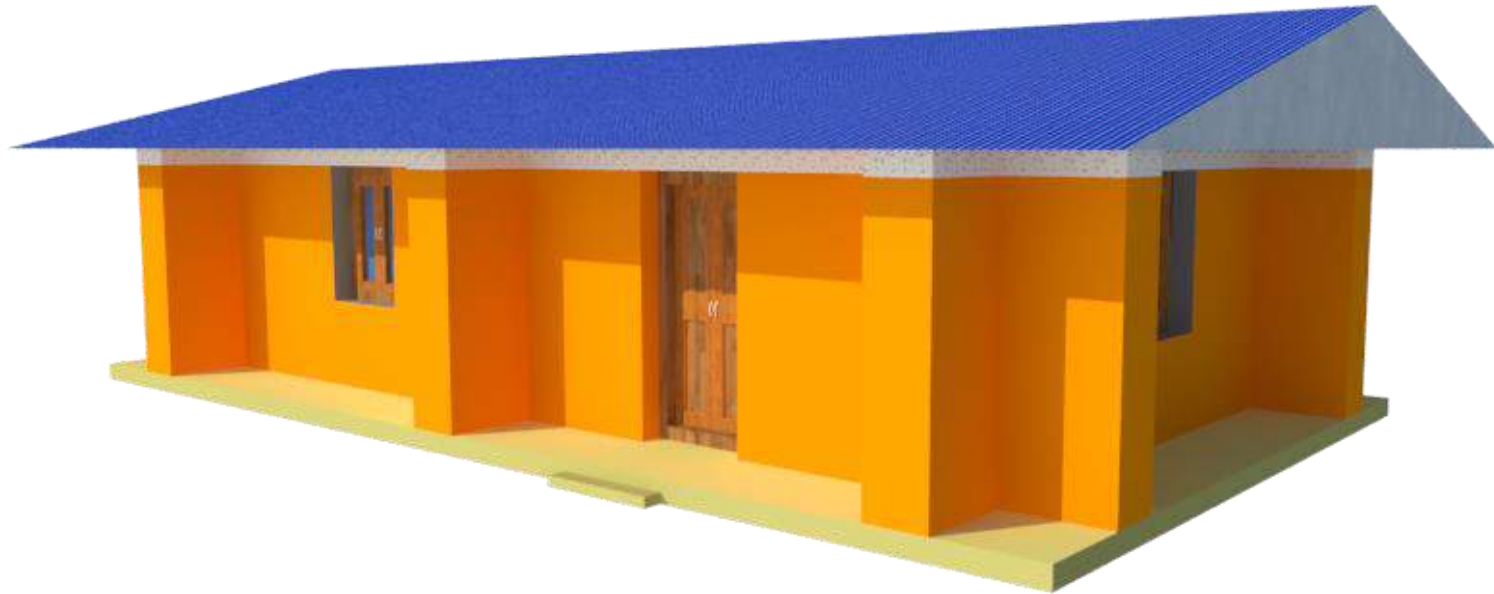
HOUSING TYPE: SECTION
DRAWING TITLE: 110

SCALE: NONE
DATE:

3/10

8. Earth Bag Masonry

- Earthbag technology is a simple, inexpensive and sustainable method for building structures using ordinary soil found at construction site. The technology consists of Polypropylene bags filled with locally available soil, laid similarly to masonry with barbed wire serving as a mortar and provides tensile as well as shear strength.
- The featured design of Earthbag technology EB 8.1 consists of single storied structure with two rooms. The wall system uses Polypropylene bags filled with soil whereas CGI sheet is used for covering the roof along with wooden rafters and purlins.
- **MATERIAL PROPERTIES**
- Soil for Earthing: 25% - 30% clay & 70% - 75% Sandy soil
- Bags: Polypropylene bags
- Barbed wire: 14guage, 4 pointed
- Rebar: Mild steel bar of Grade Fe 250
- Nominal Mix Ratio : 1:1.5:3 (C:S:A)



LEVEL	MATERIALS								
	Stone	Cement	Sand	Aggregate	Reinforcing Bar	CGI Sheet	GI Sheet	Wood	Earth
	Cu.m.	Bags	Cu.m.	Cu.m.	Kg.	Bundle	Sq.m.	Cu.m.	Cu.m.
Up to Plinth Level	12.3	-	-	-	-			-	-
Super Structure	-	9	1.0	2.0	237.3			0.5	25.5
Roofing	-	-	-	-	-	4.2	8.0	2.4	
TOTAL	12.3	9	1.0	2.0	237.3	4.2	8.0	2.9	25.5

MODEL E.B.-8.1

E.B.-8.1



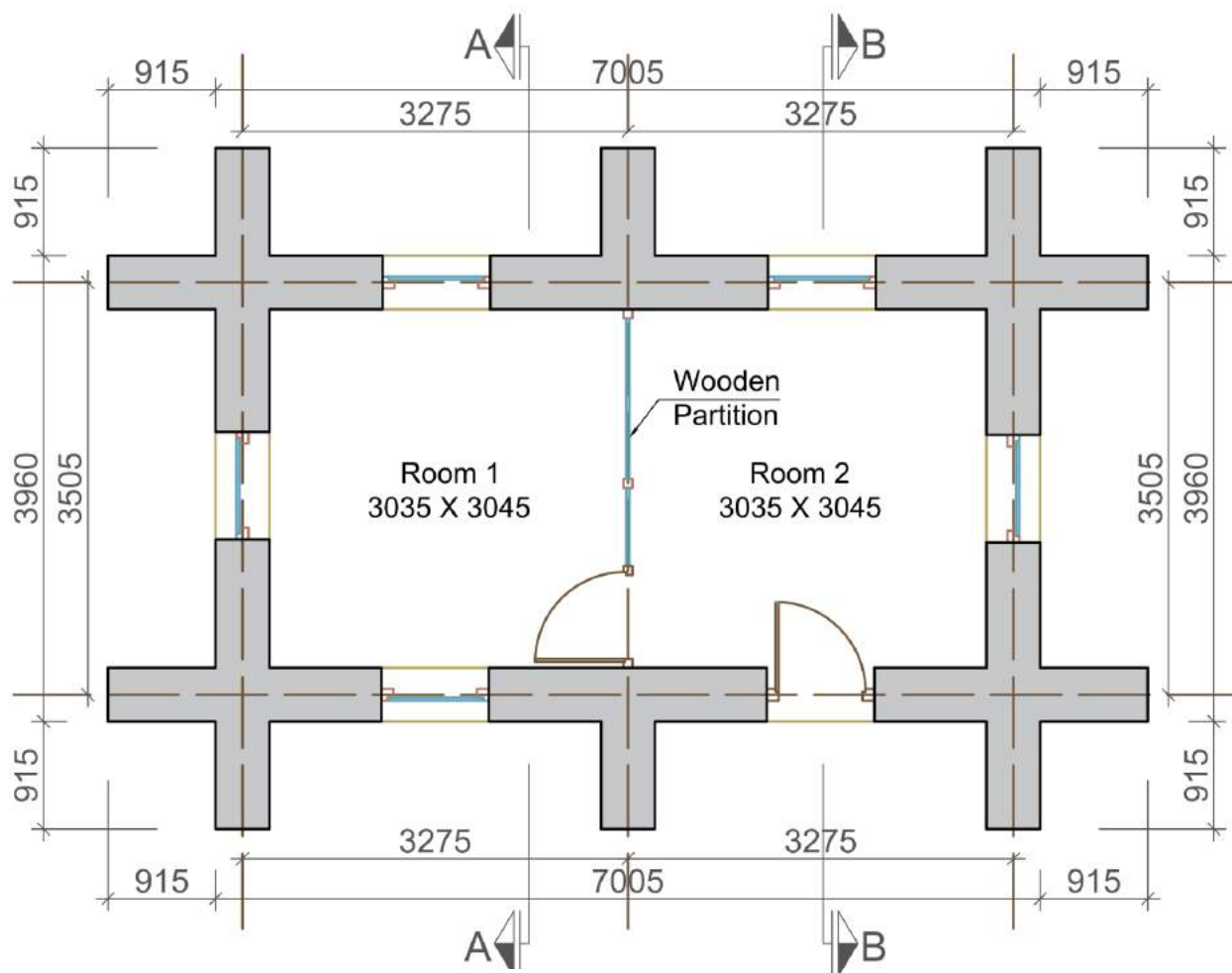
MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: ESTIMATE AND 3D-VIEW

SCALE: NONE

DRAWING TITLE: 120

DATE:

**GROUND FLOOR PLAN**

FLOOR AREA: 31.95 SQ.M.

MODEL E.B.-8.1

E.B.-8.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: FLOOR PLAN

DRAWING TITLE: 121

SCALE: NONE

DATE:

2/11



FRONT ELEVATION



SIDE ELEVATION



BACK ELEVATION



SIDE ELEVATION

MODEL E.B.-8.1

E.B.-8.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

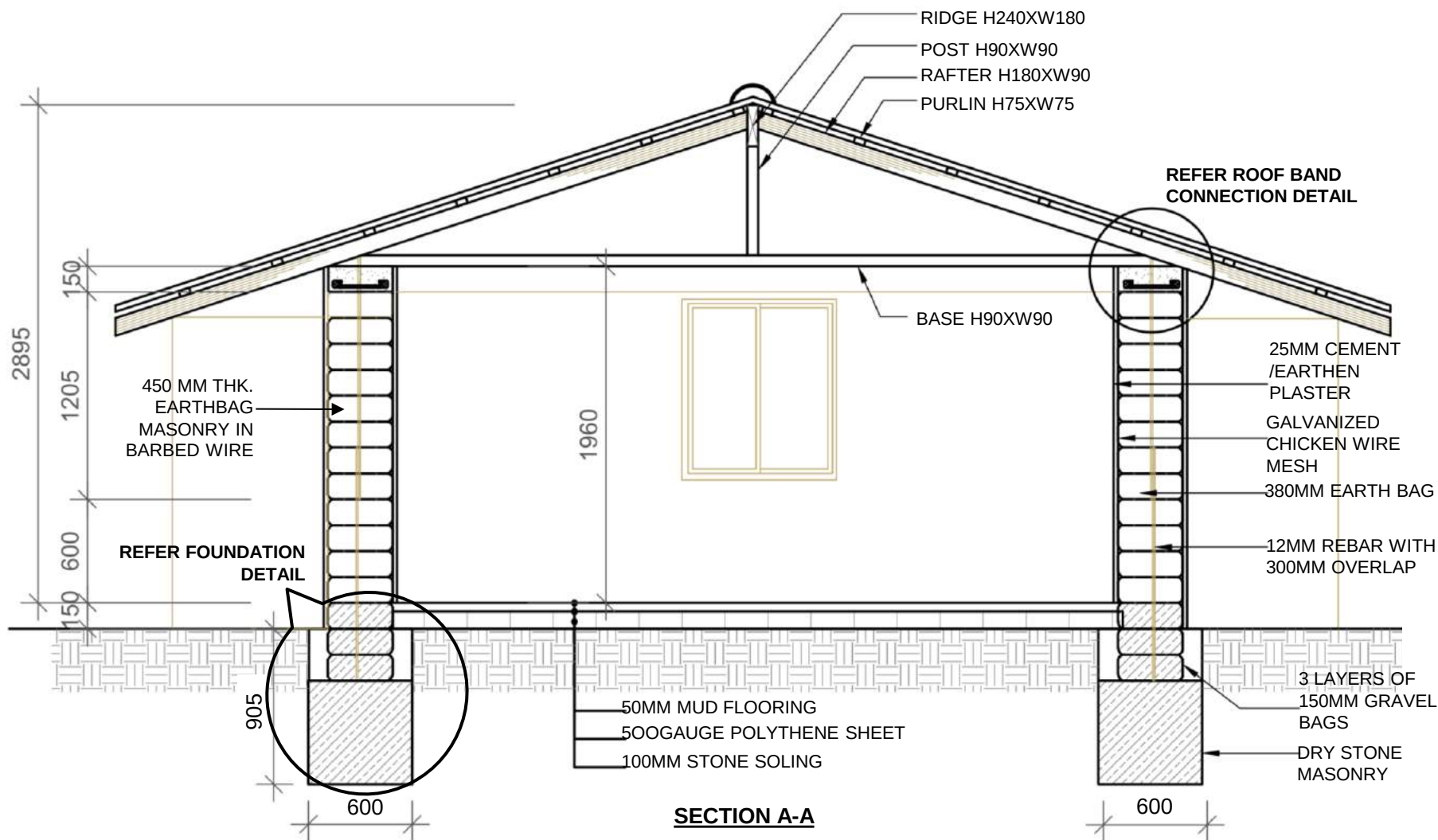
HOUSING TYPE: ELEVATIONS

SCALE: NONE

3/11

DRAWING TITLE: 122

DATE:



SECTION A-A

MODEL E.B.-8.1

E.B.-8.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: SECTION

SCALE: NONE

4/11

DRAWING TITLE: 123

DATE:

9. Light Gauge Steel Structure

- Cold Form Light gauge steel construction is a structural system consisting of thin steel sections clad with light gauge steel panel, Cellular light weight concrete, Cement fiber board, gypsum board or calcium silicate board. The steel sections used here are called *cold formed* sections, meaning that the sections are formed, or given shape at room temperature. This kind of technology requires high level of planning and precision as cold formed sections are fabricated at factory. Similarly skilled manpower are required in site for precise execution of designs.
- Featured design L.G.S 9.1 is a single storied residential unit with 2 bedrooms. Model L.G.S 9.2 is a two storied residential units with 4 bedrooms.
- **MATERIAL PROPERTIES**
- The raw materials used for the LGS steel frame is Galvanized cold form steel stripe
- Yield strength:
- Min. 450 N/mm² for LGS 9.1
- Min. 350 N/mm² for LGS-9.2
- Galvanized zinc coated: Min. 275gsm



LEVEL	MATERIALS								
	Brick	Cement	Sand	Aggregate	Reinforcing Bar	CGI Sheet	GI Sheet	MS angles & Plates	Wall Board
	No.	Bags	Cu.m.	Cu.m.	Kg.	Bundle	Sq.m.	Cu.m.	Sq.m.
Up to Plinth Level	2,973.0	87.5	6.4	9.5	594.5			-	2.5
Super Structure	-	2.5	0.3	-	-			4,184.8	244.3
Roofing	-	-	-	-	-	5.4	14.8	2,629.8	
TOTAL	2,973.0	2.5	6.7	9.5	594.5	5.4	14.8	6,814.6	246.8

MODEL L.G.S.-9.2

L.G.S.-9.2



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

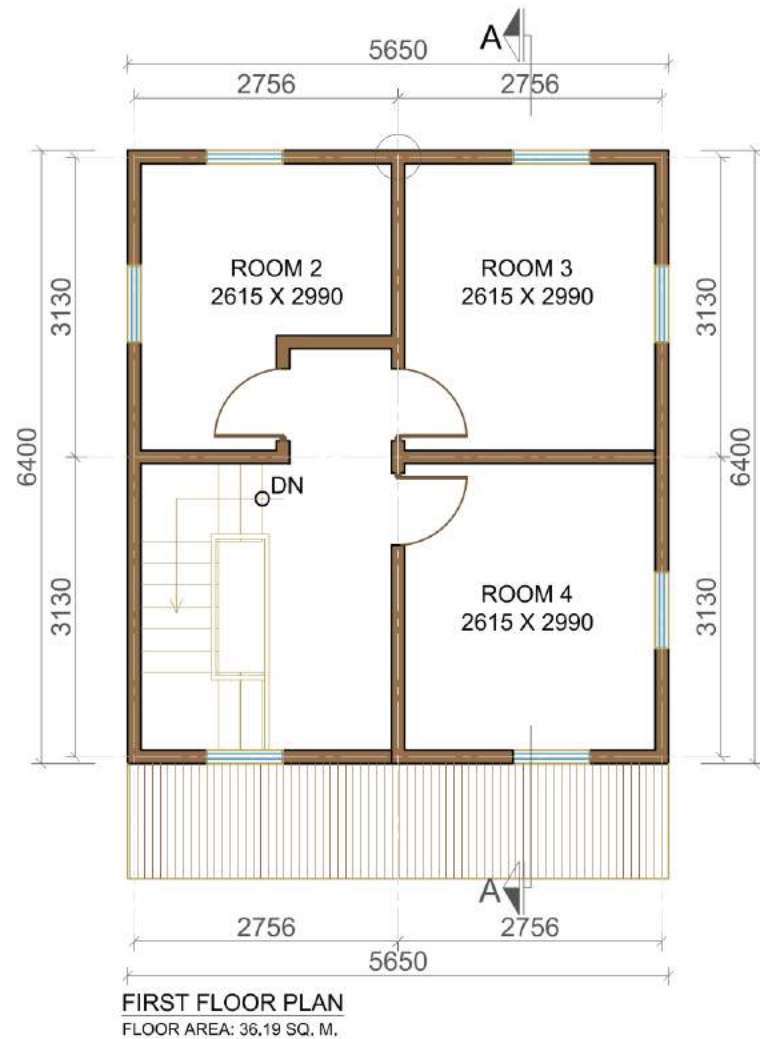
HOUSING TYPE: ESTIMATION AND 3D-VIEW

SCALE: NONE

DRAWING TITLE: 142

DATE:

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MODEL L.G.S.-9.2

L.G.S.-9.2



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

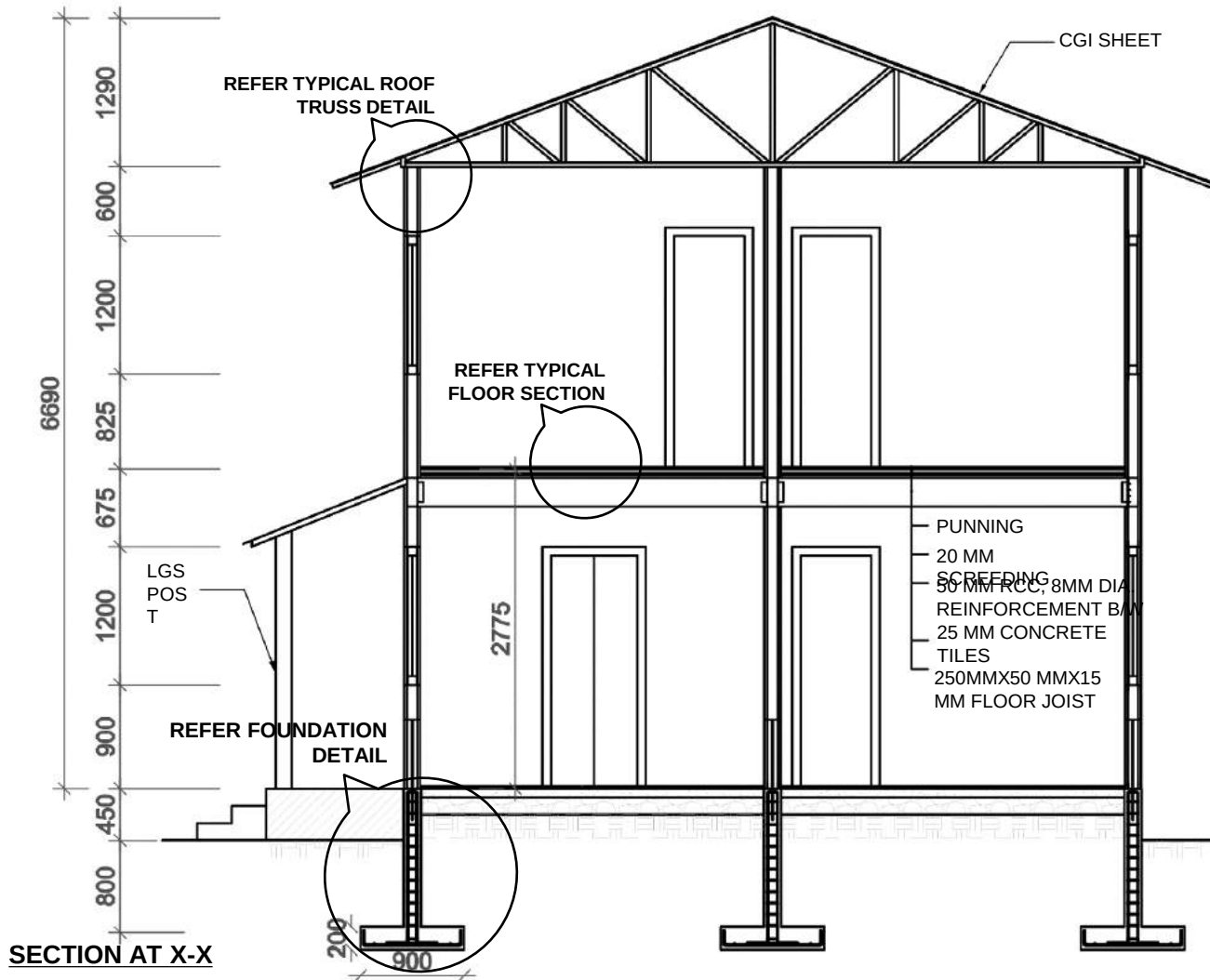
HOUSING TYPE: FLOOR PLANS

DRAWING TITLE: 143

SCALE: NONE

DATE:

2/7



MODEL L.G.S.-9.2

L.G.S.-9.2



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: SECTION

DRAWING TITLE: 145

SCALE: NONE

DATE:

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10. Steel Structure

- S.S 10.1 is a structural system consisting of mild steel columns and beams to make steel moment resisting frame system. Both the gravity and lateral load is resisted by moment resisting frame. The floor system is made of profile metal decking system over which the thin layer of RCC is laid. The roofing system consists of MS Steel tubes truss with CGI Sheet. The infill wall consists of light weight partition wall made of light weight material having density less than 1000Kg/m^3
- The featured design consists of two storey residential building consisting of 6 nos. of room.
- **MATERIAL PROPERTIES AND SPECIFICATION**
- Structural Steel Yield Strength: Fe250
- CGI Sheet: min 53 gauge
- Infill material density $\geq 1000\text{kg/m}^3$
- Mix ratio grade: 1:1.5:3
- Tensile Strength of rebar: Fe 500



LEVEL	MATERIALS										
	Brick	Cement	Sand	Aggregate	Reinforcing Bar	MS pipe	Steel sections	CGI Sheet	GI Plain sheet	Aluminium Door	Aluminium Window
	No.	Bags	Cu.m.	Cu.m.	Kg.	Kg.	Kg.	Bundle	Sq.m.	Sq.m.	Sq.m.
Up to Plinth Level	3,384.0	130.0	11.0	13.0	974.0	-	-			-	-
Super Structure	-	111.0	5.0	10.0	582.0	-	3,930.1			10.3	22.6
Roofing	-	-	-	-	-	845.4	-	6.3	11.1	-	-
TOTAL	3,384.0	241.0	16.0	23.0	1,556.0	845.4	3,930.1	6.3	11.1	10.3	22.6

MODEL S.S.-10.1

S.S.-10.1



MINISTRY OF URBAN DEVELOPMENT
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BUILDING CONSTRUCTION

HOUSING TYPE: ESTIMATION AND 3D-VIEW

SCALE: NONE

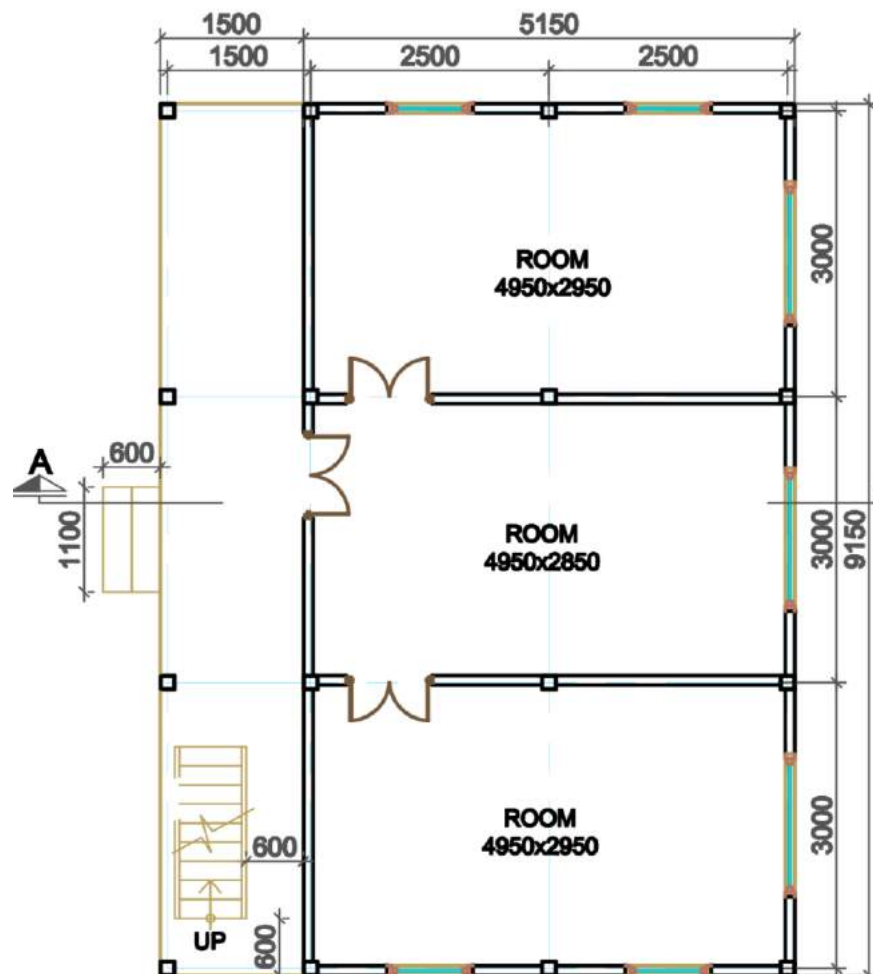
DRAWING TITLE: 151

DATE:

1/11

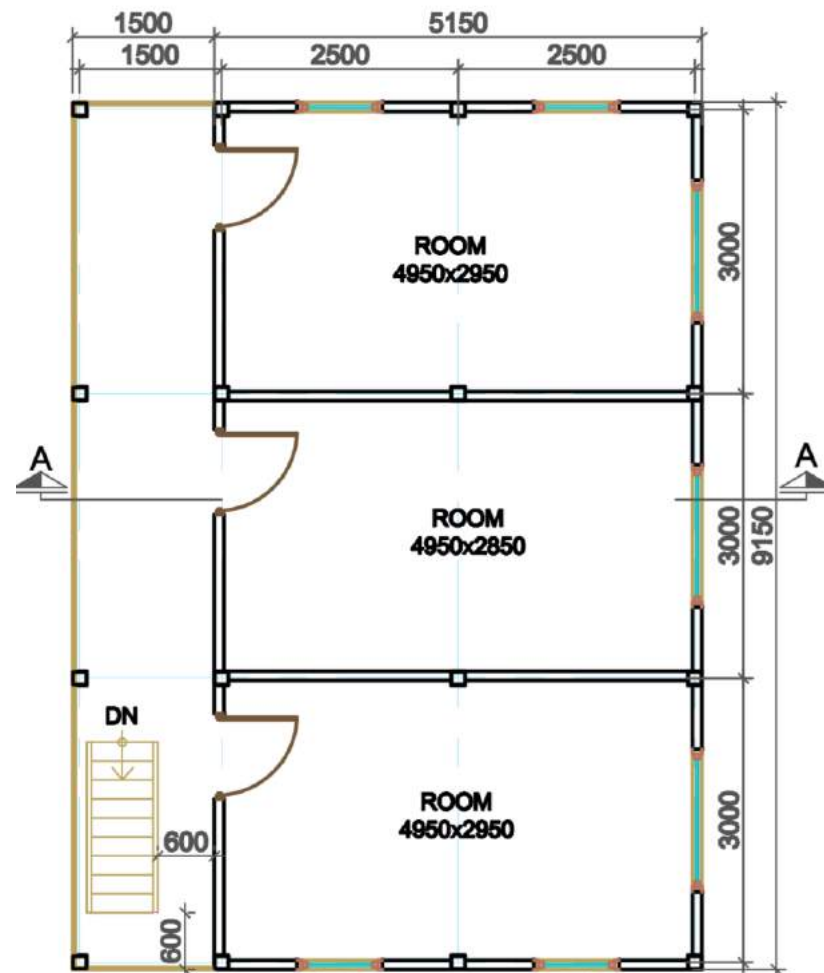
MODEL S.S.-10.1, STEEL STRUCTURE

TWO STOREY



GROUND FLOOR PLAN

AREA: 60.85 SQ.M



FIRST FLOOR PLAN

AREA: 60.85 SQ.M

MODEL S.S.-10.1

S.S.-10.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

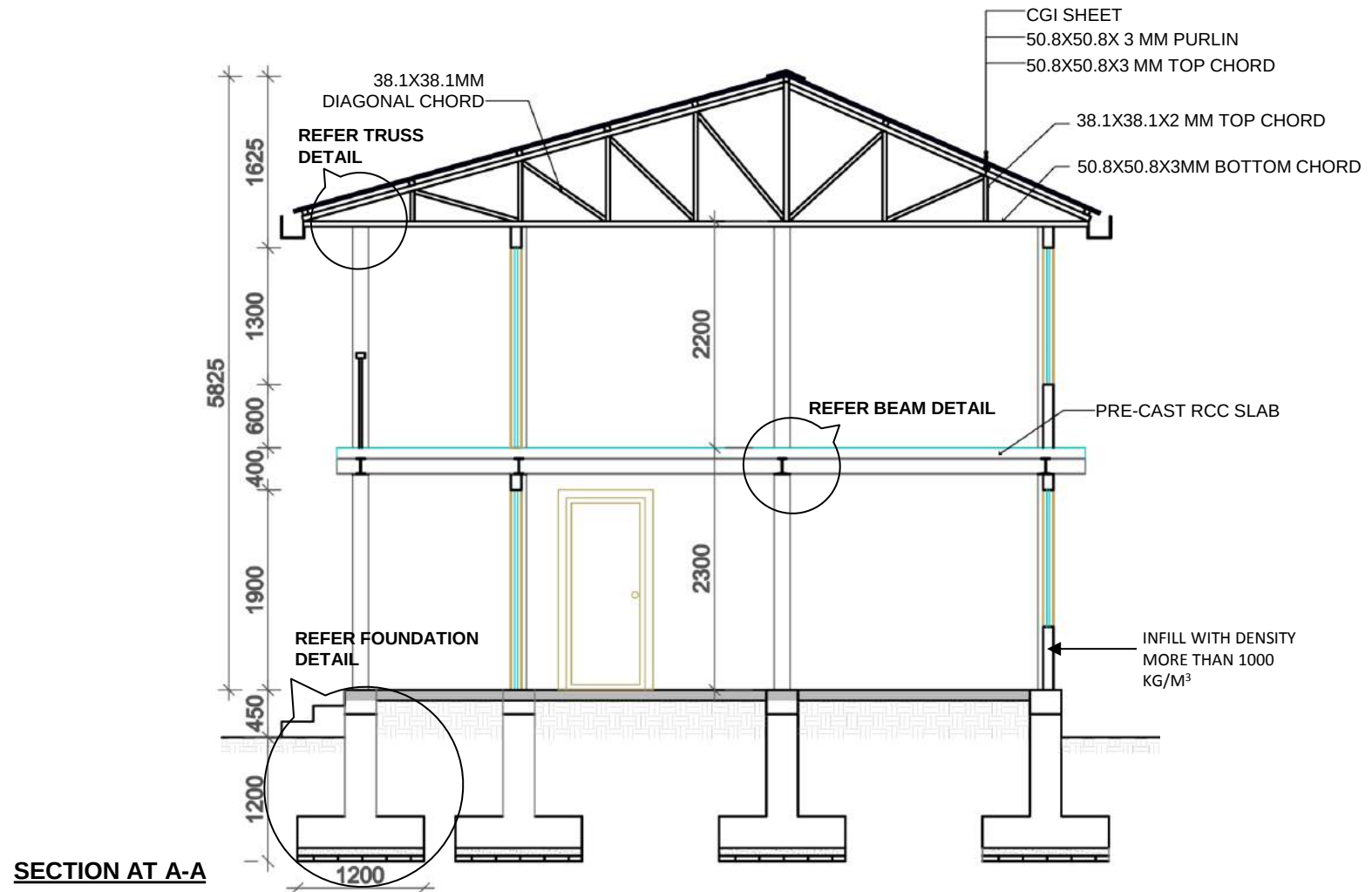
HOUSING TYPE: FLOOR PLANS

DRAWING TITLE: 152

SCALE: NONE

DATE:

2/11



MODEL S.S.-10.1

S.S.-10.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: SECTION

SCALE: NONE

DRAWING TITLE: 154

DATE:

4/11

10. Timber Structure

- T.S 11.1 is a structural system consisting of timber studs (vertical members) and horizontal member load bearing system. The gravity load is resisted by the studs and lateral load is resisted by the timber bracing located at strategic positions. The floor system consists of wooden joist over which the wooden planks are laid. The roofing system consists of wooden truss system with CGI sheet. The timber planks are used as light weight partition walls
- The featured design consists of two storied resident having 6 number of rooms.
- **PROPERTIES OF TIMBER**
- Density: 640 kg/m^3
- Modulus of Elasticity: $9.4 \times 10^3 \text{ N/mm}^2$
- Binding & tension along Grains, Extreme Fiber Stress, inside location: 13.7 N/mm^2
- Binding & tension along Grains, Extreme Fiber Stress, outside location: 11.4 N/mm^2
- Shear Stress, Horizontal in Beams all locations: 1 N/mm^2
- Shear Stress, along grains all locations: 1.4 N/mm^2
- Compressive Stress, inside location (parallel to grains): 8.6 N/mm^2
- Compressive Stress, outside location (parallel to grains): 7.7 N/mm^2



LEVEL	MATERIALS										
	Brick	Cement	Sand	Aggregate	Reinforcing Bar	MS Angle & Plates	Wood	CGI Sheet	GI Plain sheet	Aluminium Door	Aluminium Window
	No.	Bags	Cu.m.	Cu.m.	Kg.	Kg.	Cu.m.	Bundle	Sq.m.	Sq.m.	Sq.m.
Up to Plinth Level	3,652.7	113	16.7	8.7	630.0	-	-			-	-
Super Structure	-	18	1.2	2.2	-	526.3	16.3			10.3	22.6
Roofing	-	-	-	-	-	-	0.2	6.7	11.4	-	-
TOTAL	3,652.7	131	17.9	10.9	630.0	526.3	16.5	6.7	11.4	10.3	22.6

MODEL T.S.-11.1

T.S.-11.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

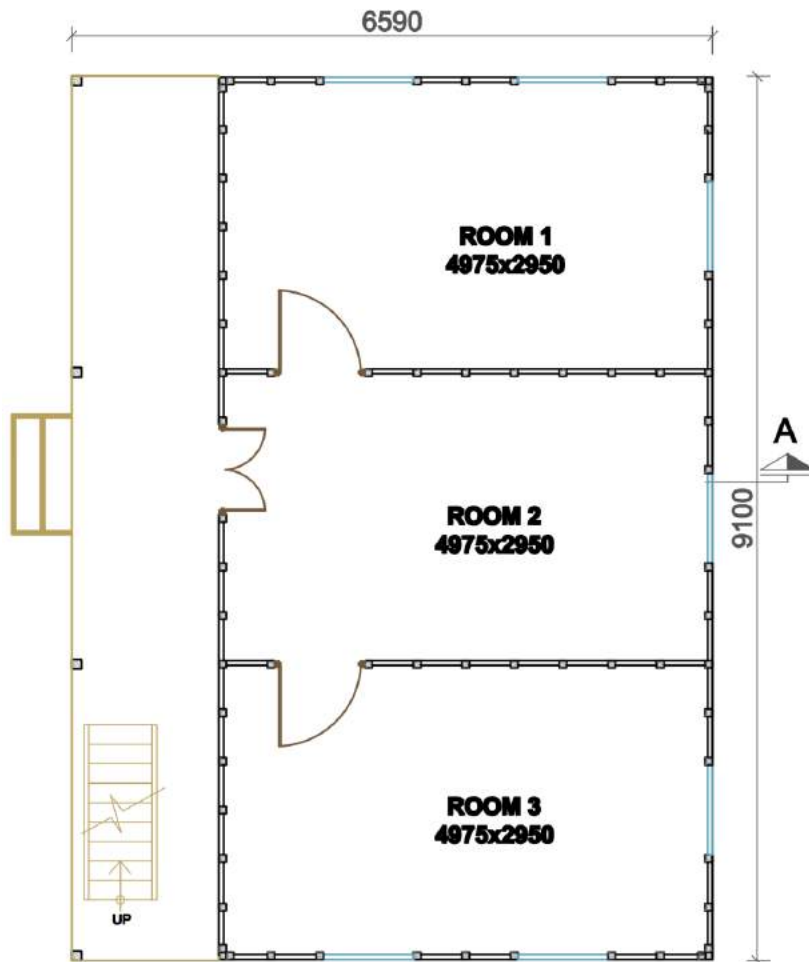
HOUSING TYPE: ESTIMATE AND 3D-VIEW

SCALE: NONE

DRAWING TITLE: 164

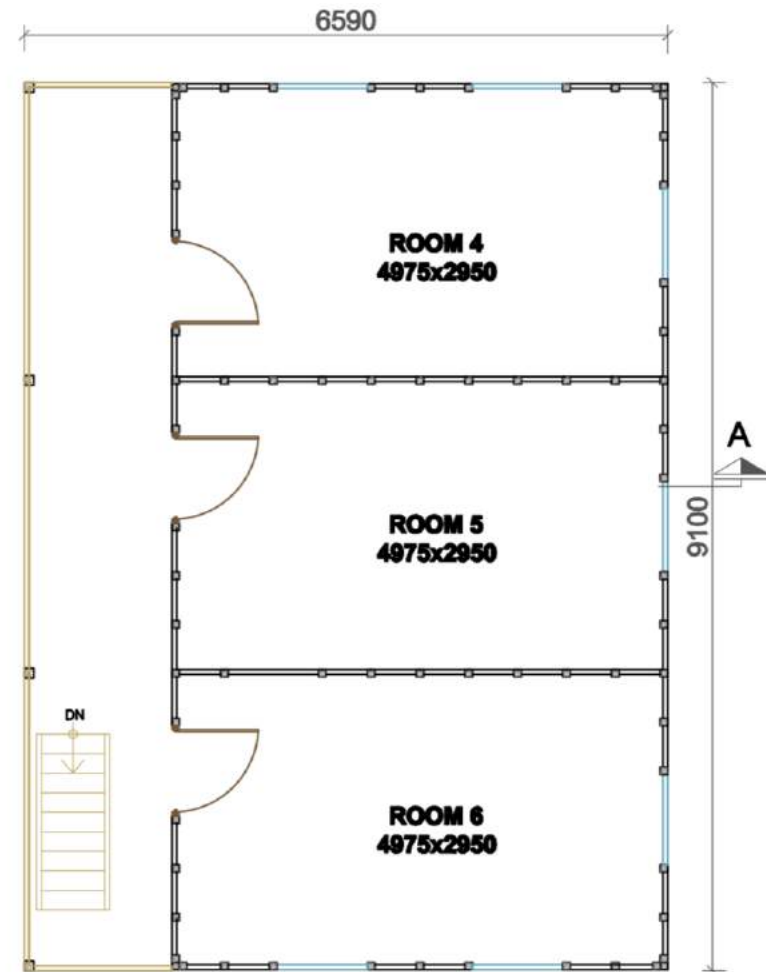
DATE:

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GROUND FLOOR PLAN

AREA: 61.64 SQ.M.



FIRST FLOOR PLAN

AREA: 61.64 SQ.M.

MODEL T.S.-11.1

T.S.-11.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

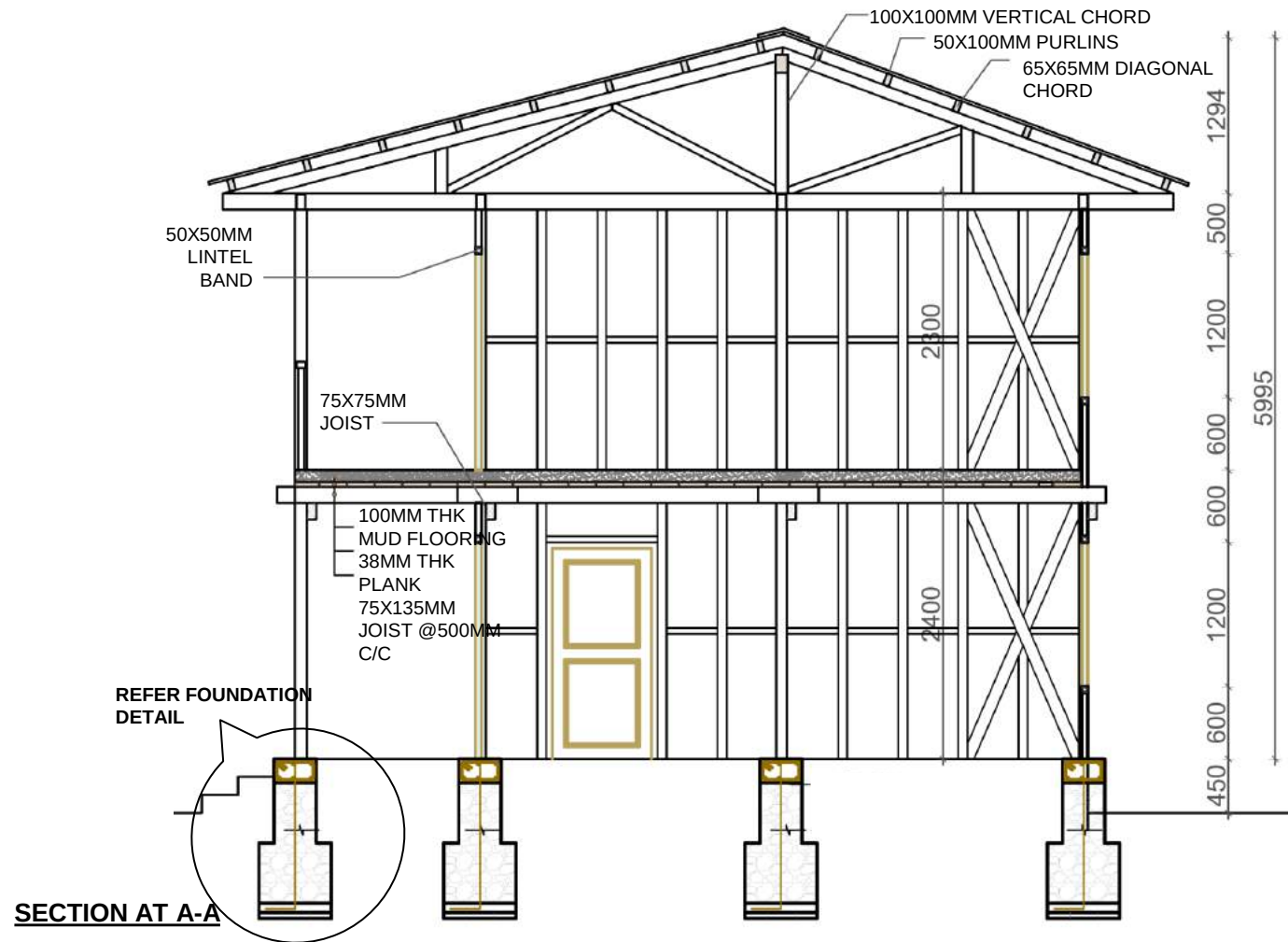
HOUSING TYPE: FLOOR PLANS

DRAWING TITLE: 165

SCALE: NONE

DATE:

2/8



MODEL T.S.-11.1

T.S.-11.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE:

SECTION

SCALE: NONE

DRAWING TITLE: 167

DATE:

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12. Debris Block Masonry

- The technology proposes residence construction with block made from stone or brick debris stabilized with cement. The objective of the design is to contribute towards resilient models that helps in debris management as well as improves safety in future earthquakes.
- Featured design D.B 12 is a single storied model house with 2 rooms. Bands are provided at plinth level, sill level, corner, lintel level and roof level. Roofing is of corrugated Galvanized Iron sheets under wooden rafters
- **MATERIAL PROPERTIES**
- **For mud mortar stone masonry**
- Size: 300 mm length × 150 mm width × 200 mm height
- Color: light grey
- Density: 2000 Kg/cm³ to 2300 Kg/cm³
- **For mud mortar brick masonry**
- Size: 300 mm length × 150 mm width × 200 mm height
- Color: light grey
- Density: 1700 Kg/cm³ to 2200 Kg/cm³



LEVEL	Debris Block	Cement	Sand	Aggregate	Reinforcing Bar	CGI Sheet	GI Sheet	Wood
	No.	Bags	Cu.m.	Cu.m.	Kg.	Bundle	Sq.m.	Cu.m.
Up to Plinth Level	1,762.0	34.2	2.4	4.7	273.5			-
Super Structure	1,923.0	21.8	1.2	2.3	307.1			0.3
Roofing	-	-	-	-	-	5.2	8.3	2.8
TOTAL	3,685.0	56	3.6	7.0	580.6	5.2	8.3	3.1

D.B. 12.1

D.B. 12.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

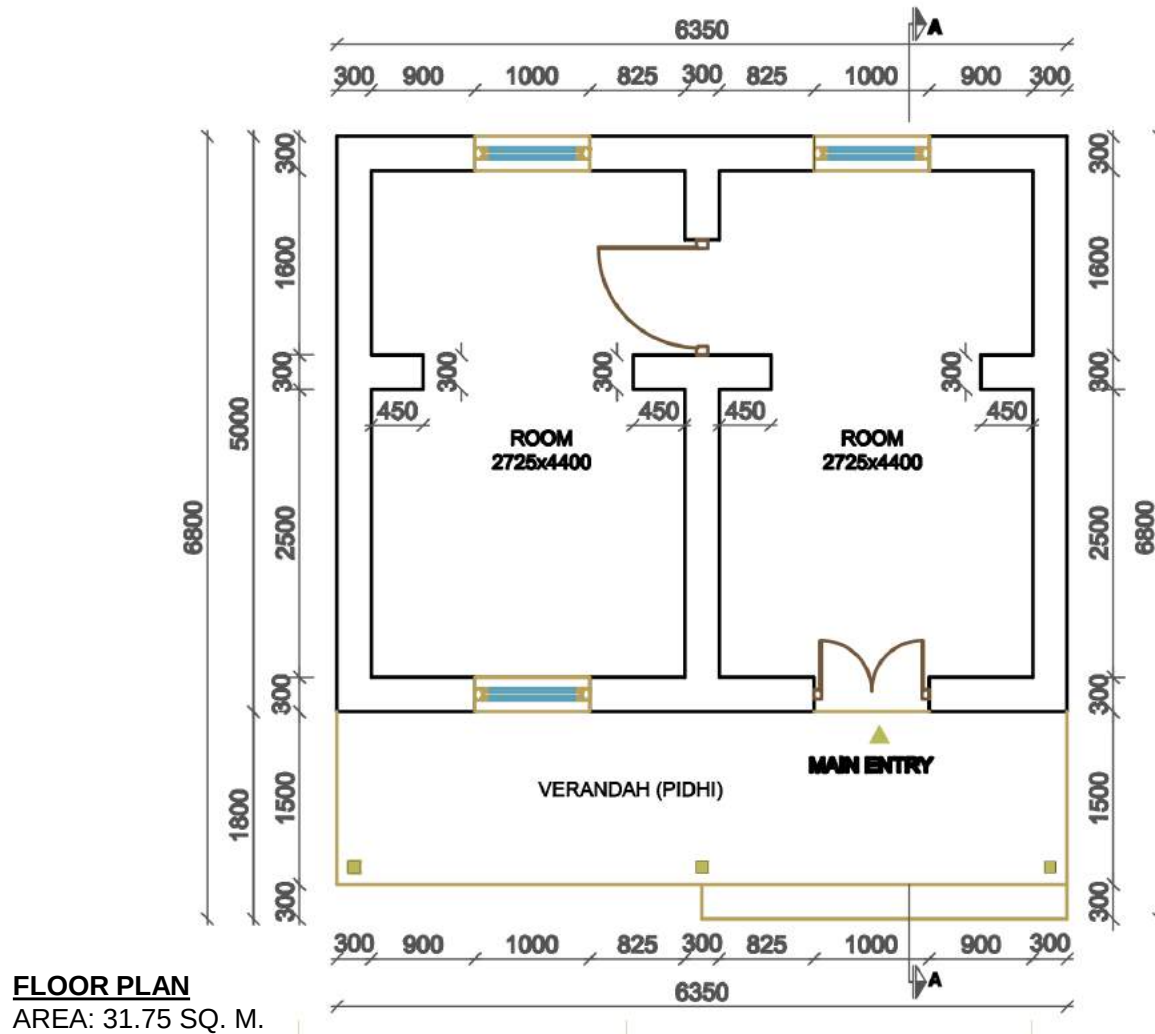
HOUSING TYPE: **ESTIMATE AND 3D-VIEW**

SCALE: NONE

DRAWING TITLE: 174

DATE:

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D.B. 12.1

D.B. 12.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

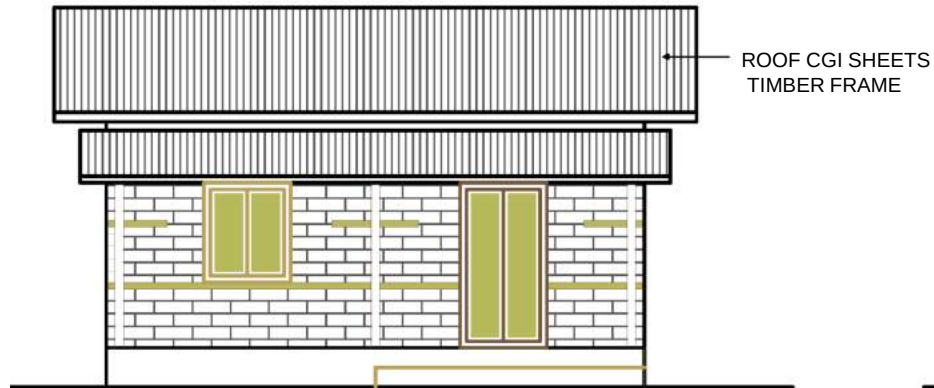
HOUSING TYPE: FLOOR PLAN

DRAWING TITLE: 175

SCALE: NONE

DATE:

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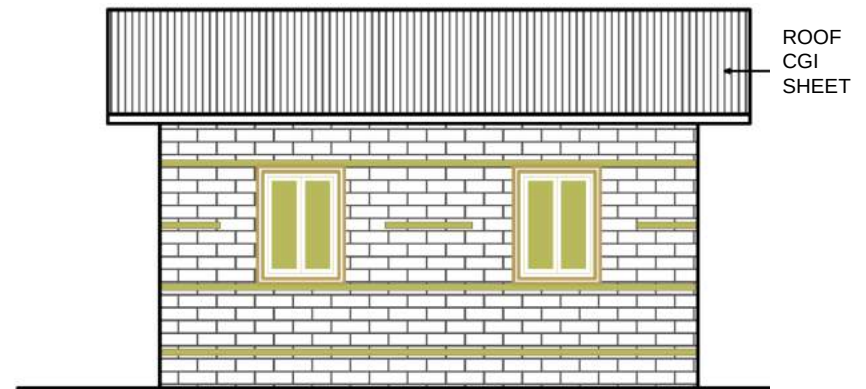
FRONT ELEVATION



SIDE ELEVATION

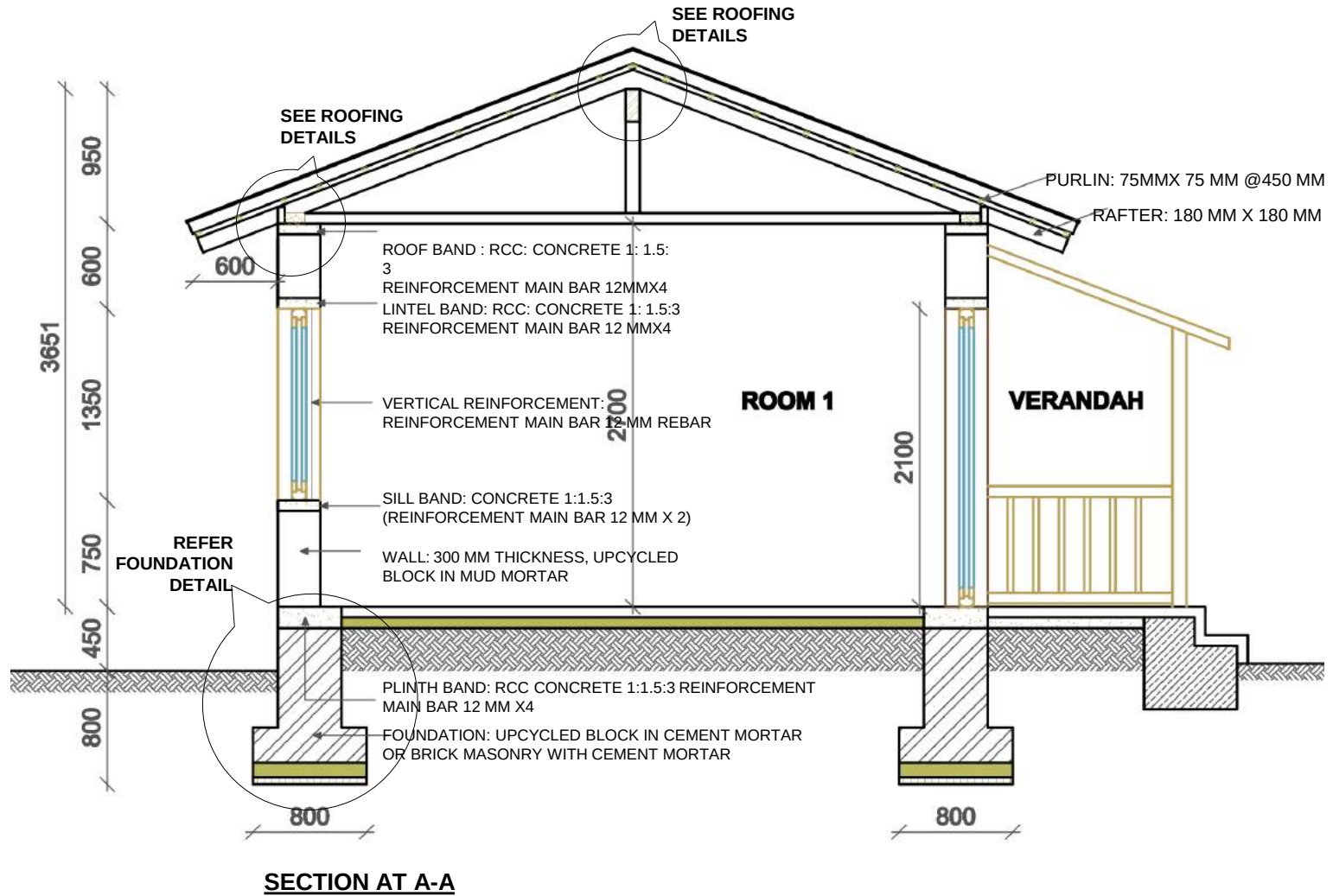


SIDE ELEVATION



BACK ELEVATION





D.B. 12.1

D.B. 12.1



MINISTRY OF URBAN DEVELOPMENT
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION

HOUSING TYPE: SECTION
DRAWING TITLE: 177

SCALE: NONE
DATE:

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Thank you